



Springback prediction in tube free bending based on simulation and WOA-Attention-BP hybrid neural network

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ABSTRACT

Accurate springback radius measurement and prediction are essential for springback compensation and forming accuracy control in tube free bending. However, conventional contact measurement methods and existing springback prediction models face challenges in balancing accuracy and efficiency. Therefore, this study proposes a metrologically characterized springback radius measurement method, and integrates analytical modeling, finite element (FE) simulation, and machine learning techniques for springback radius prediction. First, the fitted circular radius of the centerline in the constant-curvature region after unloading is defined as the measurand. A measurement chain integrating 3D scanning, point-cloud processing, centerline extraction, and two-stage circle fitting is established. The impact of coordinate perturbations on the fitted radius is quantified using a radius error amplification factor, and a complete measurement uncertainty budget is developed. For Al6061 aluminum alloy tubes, the proposed method achieves an expanded uncertainty of 0.86 mm ($k=2$) and a repeatability standard deviation of 0.63 mm. Furthermore, a hybrid WOA-Attention-BP prediction framework is developed, where WOA optimizes network parameters and attention mechanism adaptively weights input features. The model achieves prediction performance with MAE, RMSE, and MAPE of 3.13 mm, 3.52 mm, and 2.09%, respectively, outperforming other mainstream forecasting models in accuracy. Furthermore, by integrating measurement uncertainty into the prediction residuals, the model prediction uncertainty is determined to be 3.67 mm. This study provides a reliable theoretical foundation for rapid springback prediction in free bending and applications in intelligent manufacturing.

1. Introduction

Al6061 aluminum alloy tubes, as key lightweight components, have become an essential material in modern industrial systems such as aerospace, new energy vehicles, and high-end equipment manufacturing. This is attributable to their excellent processing properties, welding and electroplating characteristics, superior corrosion resistance, and high toughness, making it suitable for conveying media such as hydraulic fluid, fuel and coolant [1,2]. In recent years, the free bending technology, with its mold-free forming process flexibility, has provided a significant technical path for the lightweight and integrated manufacturing of high-performance tubular components [3–5]. However, Al6061 exhibits significant springback behavior after bending forming, and the dimensional

deviation caused by springback remains a key issue restricting the technology's expansion into high-precision and high-reliability applications [6,7]. Research shows that springback is a complex mechanical behavior resulting from the coupled effects of multiple factors such as material mechanical properties, geometric dimensions, and process parameters. This multifactorial nonlinear coupling characteristic increases the difficulty of accurately measuring and reliably predicting springback values [8–11]. In addition, factors such as measurement equipment accuracy, data processing methods, and test repeatability also affect springback measurement results and the evaluation of prediction models. Therefore, there is an urgent need to develop a fast and highly accurate method for estimating springback radius to optimize forming processes and improve the manufacturing precision of tubular components.

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Up to now, two types of methods, model-based [12–16] and data-driven [17–22], are used to estimate springback in tube bending. The model-based approach can be further divided into mathematical modeling and FE modeling methods. Among them, Ma et al. [23] proposed a generalized analytical model applicable to the springback of tube rotary draw bending. By considering material properties, geometric parameters, and key process parameters, they achieved accurate prediction of springback under the large bending angle ($\geq 90^\circ$). Zhang et al. [24] established a springback prediction formula based on the elastic-perfectly plastic material model, and the theoretical calculation showed good consistency with the experimental results. Liu et al. [25] established an analytical prediction model for the springback of profile draw bending based on the strain neutral layer theory, analyzed the influence of material and geometric parameters on springback. However, most analytical models are only applicable to conventional rotary draw bending processes and inadequately account for the dynamic evolution of the strain neutral layer during free bending and its impact on springback, making it difficult to balance computational efficiency, prediction accuracy, and process applicability. To reduce the limitations imposed by complex boundary conditions and material nonlinearity on analytical modeling, the FE method has been widely applied to the analysis of free bending mechanisms and springback prediction. Li et al. [4] carried out numerical simulation on related process parameters of bending forming, and confirmed that the FEM has significant advantages in optimizing the forming accuracy of components. Wang et al. [12] combined FE simulation with geometric analysis to investigate the influence of process parameters on springback angle, but their discussion on multi-factor interactions and uncertainty propagation remains limited. Although the FE simulation is superior to theoretical analytical methods in terms of accuracy and mechanism revelation, its high computational cost owing to the need for high-precision implicit solving limits its direct application in intelligent manufacturing processes.

In recent years, with the rapid development of artificial intelligence technology, training existing process data using machine learning methods has become a new approach for the rapid and accurate prediction of springback and process optimization [13,14]. Lu et al. [15] constructed a machine learning model based on FE simulation data to optimize the forming path of the stretch bending process. Sun et al. [16] proposed an integrated framework fusing the Gated Recurrent Unit (GRU) and the Pb-NSGA-III multi-objective optimization algorithm. They utilized FE simulation data to predict the springback angle and radius of tubes. However, the large gap between FE simulation results and experiments affected the prediction accuracy of the model. He et al. [26] integrated the springback analysis relationship into a convolutional neural network, and combined it with a real-time force measurement system and soft sensing techniques to achieve springback prediction during rotary bending processes. Liu et al. [27] developed a BP-ANN prediction model for the manufacturing process of U-shaped thin-walled metal tubes, aiming to predict optimal bending radius and bending angle, and validated the model through FE simulation and experiments. Among these machine learning models, the shallow architecture of the BP neural network gives it a lower risk of overfitting compared to deep models (GRU), particularly under small sample conditions. However, the traditional BP neural network is limited in its application potential due to its low generalization ability and tendency to get trapped in local optima. To address this, scholars have begun to explore hybrid modeling strategies of optimization algorithms and neural networks. Xu et al. [28] enhanced the global search capability of BPNN by employing genetic algorithms (GA) and an improved particle swarm optimization (PSO) algorithm, thereby improving the model's convergence performance and prediction accuracy. Wasif et al. [29] conducted multi-objective optimization of process parameters using V-bending tests and genetic algorithms. Although traditional optimization algorithms such as GA and PSO overcome the BP network's high sensitivity to initial weights by introducing a global search mechanism, their performance is highly dependent on the setting of their own hyperparameters, and the complex

parameter tuning process increases the uncertainty of the model. To address this challenge, the whale optimization algorithm (WOA) proposed by Mirjalili et al. [30] achieved a dynamic balance between global exploration and local exploitation through its unique spiral hunting mechanism and convergence characteristics, demonstrating a stronger ability to escape local optima compared to GA and PSO. Then, Qin et al. [20] combined an improved whale optimization algorithm with a BP neural network for predicting alloy flow stress, and the results showed that the optimized model exhibited better prediction performance. Gao et al. [21] developed a WOA-BP neural network model for lithium-ion battery state-of-health estimation, demonstrating strong application potential. Niu et al. [22] effectively addressed measurement errors caused by dynamic sensor angles through the synergistic integration of WOA optimization and BPNN nonlinear modeling.

In addition to optimizing network parameters, differential processing of input features is also crucial for improving the reliability of springback prediction. If the model fails to effectively identify key features, weakly correlated variables and their measurement fluctuations may interfere with network training, thereby reducing the model's predictive stability and noise resistance. To address this, Chang et al. [31] introduced an attention mechanism into a hybrid model combining optimization algorithms and neural networks, dynamically adjusting the weights of different input features to enhance the model's predictive performance. Wang et al. [32] further constructed a new BP neural network framework based on the attention mechanism (BPAM++), which not only reduces computational storage costs but also alleviates the overfitting problem caused by insufficient data. These studies show that optimization algorithms and attention mechanisms can respectively improve the backpropagation neural network by optimizing network parameters and weighting input features, but their synergistic effect in predicting free bending springback of tubes has not been sufficiently investigated.

In summary, existing research on tube bending springback still has the following shortcomings:

- (1) The springback radius is commonly measured by conventional contact methods, where cross-sectional distortion and operator-induced errors may affect measurement accuracy, and a measurement approaches with explicit error characterization and good repeatability remains absent.
- (2) Most analytical models fail to adequately account for the dynamic evolution of the strain neutral layer during free bending, thereby limiting the applicability of springback models.
- (3) There is a lack of a unified data fusion and uncertainty evaluation framework linking FE simulations, experimental measurements, and machine learning predictions, making it difficult to simultaneously achieve global optimization of network parameters and adaptive identification of critical input features under small sample conditions.

To address the above issues, this study takes the bending radius of the geometric center layer after unloading in free bending of Al6061 tubes as the target measurement, and proposes a springback radius measurement method with metrological characterization. A measurement traceability chain incorporating error analysis and uncertainty evaluation is established. Furthermore, theoretical analysis and FE simulations are employed to investigate the influence of material, geometric, and process parameters on springback behavior. A hybrid WOA-Attention-BP prediction framework integrating the whale optimization algorithm (WOA), attention mechanism, and backpropagation neural network is proposed to improve the accuracy and stability of springback radius prediction.

The remainder of this paper is organized as follows. Section 2 presents the proposed measurement and estimation methodologies, including the springback radius measurement method, circle fitting and uncertainty evaluation procedures, analytical springback model, and WOA-Attention-BP prediction framework. Section 3 validates the accuracy and stability of the proposed measurement method, analyzes the effects of material properties, tube dimensions, and process parameters

on springback behavior, and evaluates the prediction performance of the WOA-Attention-BP model. Finally, Section 4 summarizes the main conclusions and discusses future research directions for tube bending springback prediction.

2. Methodology

2.1. Analysis of free bending process

Fig. 1(a) shows the principle of tube free bending. In the bending process, the tube is subjected to the bending force P_U applied by the bending die and the axial thrust P_L applied by the feed mechanism. The vertical distance of the bending die center relative to the fixed die center is called the eccentricity U . The horizontal distance between the center of the bending die and the center of the fixed die outlet is called the spacing L , which is constant. Furthermore, the bending moment can be expressed as:

$$M = P_U \times A + P_L \times U \quad (1)$$

As shown in Fig. 1(b), the tube undergoes three stages during the bending forming process: transition section C_1 , stable section C_2 and transition section C_3 . Among them, transition sections are formed by the

movement of the bending die, resulting in variable curvature arcs. When the bending die reaches the target position and remains stationary, the tube continues to advance, forming a constant curvature arc section C_2 with the bending angle depending on the feed length. Based on the geometric relationships illustrated in Fig. 1(a), the bending radius R can be calculated as:

$$R = \frac{U^2 + A^2}{2U} \quad (2)$$

2.2. Measurement of springback radius

After tube bending, springback causes a deviation between the intended geometry and the final formed geometry. Therefore, accurate measurement of the springback radius is the key to precise compensation and optimization. Based on the forming principle of free bending, the measurand is defined as the radius of the circle fitted to the centerline of the constant-curvature section C_2 after unloading. The measurement chain includes the following steps: (1) calibrating the 3D laser scanner using a calibration plate; (2) performing full-scale and multi-angle scanning of the bent component; (3) point-cloud registration and outlier removal; (4) reconstructing the surface model and extracting the tube centerline in

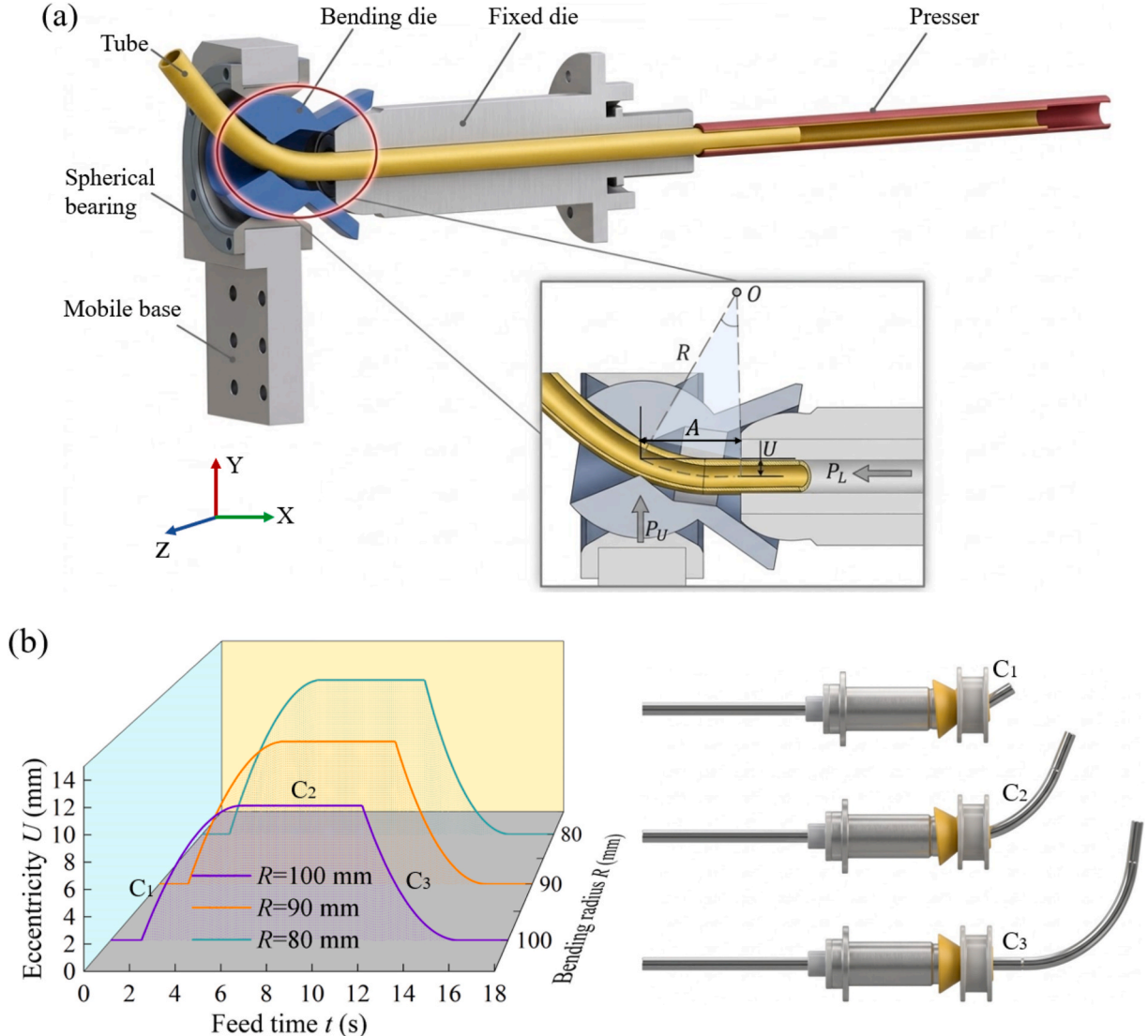


Fig. 1. Free bending forming principle and process of the tube: (a) Free bending forming principle; (b) Free bending forming process.

Geomagic Design X; (5) identifying and extracting the constant-curvature section in Geomagic Control X; (6) sampling the centerline coordinates and applying least-squares circle fitting to obtain the springback radius. Therefore, the accuracy of the fitted circle directly affects the springback radius. To obtain the optimal circle fitting, the Kasa method is first used for a fast initial fit, and then the result is further optimized using the geometric distance. Each component is independently scanned three times under the same conditions to assess repeatability.

2.2.1. Springback radius fitting process

First, suppose that N points are collected on the centerline extracted in the bending plane, with coordinates (x_i, y_i) , $i = 1, 2, 3, \dots, N$. Further, the theoretical geometric model can be expressed as:

$$(x - a)^2 + (y - b)^2 = R^2 \quad (3)$$

where (a, b) is the circle center, and R is the radius.

The circle equation can be further transformed into Eq. (4).

$$x^2 + y^2 + Bx + Cy + D = 0 \quad (4)$$

where $B = -2a, C = -2b, D = a^2 + b^2 - R^2$.

Since the actually extracted centerline sampling points are affected by image resolution, centerline extraction error, and local geometric disturbances, the circle center and radius are estimated by geometric distance least-squares based on the discrete sampling points. First, the Kasa algebraic circle fitting method is used to obtain the initial estimates of the circle center and radius, providing initial parameters for the subsequent Gauss-Newton optimization [33,34]. Further, the sum of squared algebraic distance residuals is minimized by the least-squares method, as shown in Eq. (5). Taking the partial derivatives with respect to B, C , and D and setting them to zero yields a system of linear equations as shown in Eq. (6).

$$F(B, C, D) = \sum_{i=1}^N (x_i^2 + y_i^2 + Bx_i + Cy_i + D)^2 \quad (5)$$

$$\begin{bmatrix} \sum x_i^2 & \sum x_i y_i & \sum x_i \\ \sum x_i y_i & \sum y_i^2 & \sum y_i \\ \sum x_i & \sum y_i & N \end{bmatrix} \begin{bmatrix} B \\ C \\ D \end{bmatrix} = - \begin{bmatrix} \sum (x_i^3 + x_i y_i^2) \\ \sum (x_i^2 y_i + y_i^3) \\ \sum (x_i^2 + y_i^2) \end{bmatrix} \quad (6)$$

By solving the above equations, the initial estimates of the circle center and radius, (a_0, b_0) and R_0 , can be obtained inversely from $(a, b) = (-B/2, -C/2)$ and $R = \sqrt{(B^2 + C^2)/4 - D}$.

Considering that algebraic fitting may involve a certain systematic bias in the presence of measurement error and uneven distribution of sampling points, the radial distance from each sampling point to the fitted circle is introduced as the geometric optimization objective on the basis of the Kasa initial fitted result, in order to further improve the geometric consistency of the fitted result. The sum of squared radial distance residuals of all sampling points is taken as the objective function, as shown in Eq. (7), and the Gauss-Newton method is used to iteratively optimize the circle center and radius, as shown in Eqs. (8) and (9).

$$\begin{cases} E_k = \sum_{i=1}^N (e_i^k)^2 = \sum_{i=1}^N (r_i^k - R_k)^2 \\ r_i^k = \sqrt{(x_i - a_k)^2 + (y_i - b_k)^2} \end{cases} \quad (7)$$

$$\begin{cases} \mathbf{P}_k = [a_k, b_k, R_k]^T \\ \mathbf{P}_{k+1} = \mathbf{P}_k + \Delta \mathbf{P}_k \\ (J_k^T J_k) \Delta \mathbf{P}_k = -J_k^T e_k \end{cases} \quad (8)$$

$$\begin{cases} e_k = [e_1^k, e_2^k, \dots, e_N^k]^T \\ J_i^k = \left[\frac{(a_k - x_i)}{r_i^k}, \frac{(b_k - y_i)}{r_i^k}, -1 \right] \\ J_k = [J_1^k, J_2^k, \dots, J_N^k]^T \end{cases} \quad (9)$$

where E_k is the sum of squared fitting residuals at the k -th iteration, e_i^k is the radial distance residual of the i -th point, r_i^k is the distance from the i -th point to the circle center, a_k, b_k , and R_k are the center x -coordinate, center y -coordinate, and fitted radius obtained at the k -th iteration, respectively. $\mathbf{P}_k$ denotes the circle parameter vector at the k -th iteration, $\Delta \mathbf{P}_k$ denotes the parameter correction at the k -th iteration, and J_i^k is the Jacobian matrix of the residual function with respect to the circle parameters.

To determine whether the iterative process has converged, the relative change of the parameters and the relative change of the objective function are used as a dual convergence criterion. Convergence is considered achieved when both values fall below 10^{-6} . Additionally, a maximum number of iterations $k_{\max} = 100$ is set. The optimized fitting radius R_m is ultimately obtained. The specific fitting process is illustrated in Fig. 2.

$$\begin{cases} \eta_\theta = \frac{\|\Delta \mathbf{P}_k\|_2}{\|\mathbf{P}_k\|_2 + \epsilon} < \epsilon_\theta \\ \eta_E = \frac{|E_{k+1} - E_k|}{E_k + \epsilon} < \epsilon_E \end{cases} \quad (10)$$

2.2.2. Evaluation of measurement uncertainty

Since the fitting radius R obtained is through nonlinear circle fitting of the centerline coordinates acquired by scanning, its uncertainty can be evaluated by propagating the coordinate errors. The fitting radius R is regarded as a function of the sampled coordinates.

$$R = f(x_1, y_1, \dots, x_N, y_N) \quad (11)$$

According to the first-order error propagation theory [35], the standard uncertainty of the fitting radius can be expressed as:

$$u^2(R) = \sum_{i=1}^N \left[\left(\frac{\partial R}{\partial x_i} \right)^2 u^2(x_i) + \left(\frac{\partial R}{\partial y_i} \right)^2 u^2(y_i) \right] \quad (12)$$

where $u(x_i)$ and $u(y_i)$ are the uncertainties of the coordinates of the i -th centerline sampling point, and $\partial R / \partial x_i$ and $\partial R / \partial y_i$ are the sensitivity coefficients of the radius to the coordinate perturbations, which can be computed by central difference.

Furthermore, a radius error amplification factor $A(\phi)$ is introduced to characterize the propagation of coordinate errors into radius errors under different fitted-arc angles.

$$A(\phi) = \frac{\sigma_R}{\sigma_{coord}} \quad (13)$$

where ϕ denotes the fitted arc angle, σ_R denotes the standard deviation of the fitted radius, and σ_{coord} denotes the standard deviation of the coordinate measurement.

To further verify the theoretical error propagation results, the Monte Carlo method is used to statistically analyze the uncertainty of the fitted radius. The standard deviation σ_R of the obtained fitted radius is used to characterize its uncertainty.

$$\sigma_R = u(R) \quad (14)$$

Considering that there are multiple error sources in the actual measurement process, the uncertainty of the final measurement result is determined by both Type A and Type B uncertainties. Type A

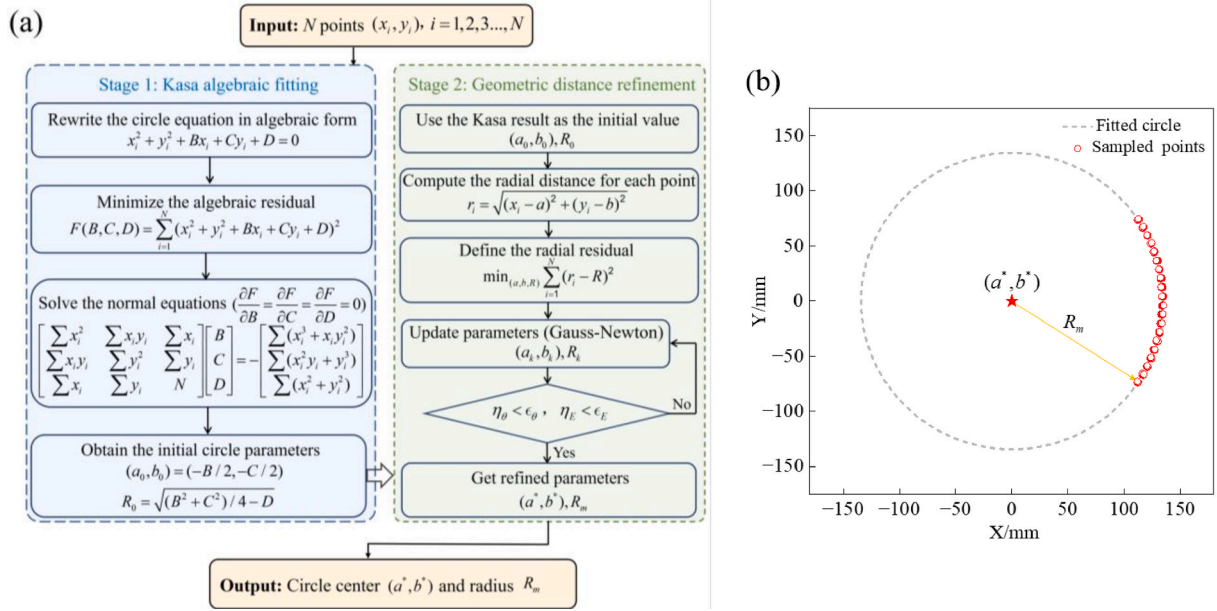


Fig. 2. Radius measurement fitting method: (a) Flowchart of the two-stage circle fitting algorithm based on Kasa algebraic initial values and geometric distance optimization; (b) Schematic illustration of least-squares circle fitting for extracted coordinates of equal curvature sections.

uncertainty mainly originates from repeated scanning experiments, while Type B uncertainty mainly includes scanner calibration error, centerline extraction, circle fitting residual, and sampling interval error. Finally, the combined standard uncertainty of the measured radius can be obtained by Eq. (16).

$$u_A = \frac{S_R}{\sqrt{n_M}} \quad (15)$$

$$u_C = \sqrt{u_A^2 + u_{B1}^2 + u_{B2}^2 + u_{B3}^2 + u_{B4}^2} \quad (16)$$

where S_R is the standard deviation of the results of repeated measurements, and n_M is the number of repeated scanning.

The final expanded uncertainty can be expressed as:

$$U_F = k u_C \quad (17)$$

where k is the coverage factor. When the confidence level is approximately 95%, $k = 2$.

2.3. Analytical springback model

To investigate the material and process parameters that affect the springback radius of tubes, this section describes the springback of tubes before and after bending based on the elastoplastic mechanics theory and clearly identifies the physical quantities that affect springback. Based on the classical elastoplastic bending theory, the following basic assumptions are introduced [36,37]:

(1) During the bending deformation process, the cross-section of the tube always remains planar, which is perpendicular to the geometric neutral axis.

(2) Friction is neglected during bending, and all material fibers are subjected to uniaxial tensile or compressive strain states.

(3) The bending deformation follows the principle of constant volume and the total strain theory.

(4) The material is a homogeneous, continuous and isotropic deformable body, and follows the loading and unloading laws of classical elastic-plastic theory.

Since the outer material of the bending is stretched under tensile stress and the inner material is compressed under compressive stress,

there exists a fiber layer at the boundary between these stresses where strain is zero, which is called the neutral layer. Owing to the hardening effect of the material, the stress distribution becomes asymmetric between the inner and outer sides, causing the neutral layer to deviate from the geometric center [37,38]. Therefore, based on the different stress-strain distributions on the cross-section during tube bending, the cross-section can be divided into different elastic deformation zones and plastic deformation zones.

As shown in Fig. 3(a), the inner radius of the tube is c and the outer radius is d , and the xOy coordinate system can be established based on the bending plane. As shown in Fig. 3(b), after the tube is bent, the neutral layer shifts. The bending radius of the neutral layer is ρ_0 , and the bending radius of the geometric center layer is R . In the local coordinate system $x_1O_1y_1$ established based on the strain neutral layer, the strain at any point P on the tube cross-section can be obtained as:

$$\varepsilon = \frac{s}{\rho_0}, \quad R - \rho_0 - d \leq s \leq R - \rho_0 + d \quad (18)$$

The bending strain can be transformed into the coordinate system xOy of the geometric center layer, and the strain can be further expressed as:

$$\varepsilon = \frac{(R - \rho_0) + y}{\rho_0}, \quad -d \leq y \leq d \quad (19)$$

According to assumptions, based on Hooke's law and the Swift hardening model, the stress at any point on the tube cross-section can be calculated by Eq. (20).

$$\sigma = \begin{cases} K \left(\frac{R - \rho_0 + y}{\rho_0} + b \right)^n, & C_4 \leq y \leq d \\ E \varepsilon, & C_5 \leq y \leq C_4 \\ -K \left(-\frac{R - \rho_0 + y}{\rho_0} + b \right)^n, & -d \leq y \leq C_5 \end{cases} \quad (20)$$

where y is the distance of any point on the tube section from the geometric center layer, K is the strength coefficient, n is the hardening exponent and E is Young's modulus.

Elastoplastic boundaries exist on both the outer and inner sides of the bending. C_4 denotes the distance from the elastic-plastic boundary on

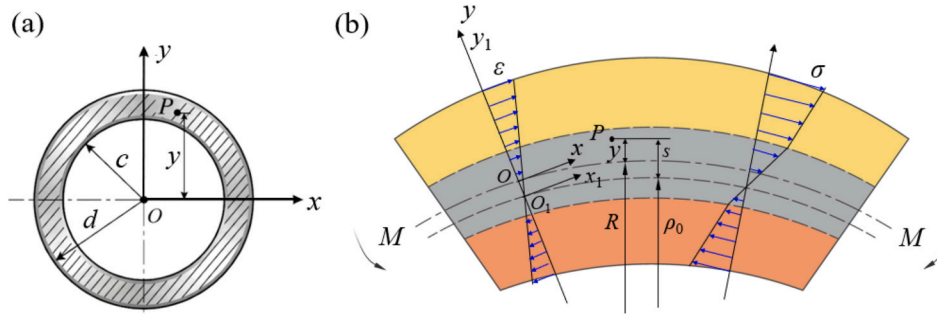


Fig. 3. Schematic diagram of tube dimensions and stress-strain distribution: (a) Tube cross-sectional dimensions; (b) The stress-strain distribution state after bending.

the outer side to the geometric center layer, and C_5 denotes the distance from the elastic-plastic boundary on the inner side to the geometric center layer. According to the elastic-plastic boundary, the bent tube section can be specifically divided into three deformation zones: tensile plastic deformation zone, compressive plastic deformation zone and elastic deformation zone. Under different bending radii, the elastic-plastic boundary can occur in two possible positions. As shown in Fig. 4 (a), the elastic-plastic boundary is located between the inner radius and the outer radius. As shown in Fig. 4(b), the elastic-plastic boundary is located between the inner radius and the geometric center layer (x-axis). To further describe the stress and strain distribution across the tube cross-section, it is necessary to establish the relationship among the elastic-plastic boundary position, the neutral layer position and the bending moment.

On the cross-section of the bent tube, the areas of the tensile plastic deformation zone, elastic deformation zone and compressive plastic deformation zone are A_1 , A_2 and A_3 respectively, the static moments about the x-axis are S_1 , S_2 and S_3 respectively, and the moments of inertia about the x-axis are I_1 , I_2 and I_3 respectively.

When the elastic-plastic boundary is located between c and d , as shown in Fig. 4(a), the areas A_1 , A_2 and A_3 of different deformation zones can be calculated by Eq. (21).

$$\begin{cases} A_1 = \int_{C_4}^d 2\sqrt{d^2 - y^2} dy \\ A_2 = \pi(d^2 - c^2) - A_1 - A_3 \\ A_3 = \int_{-d}^{C_5} 2\sqrt{d^2 - y^2} dy \end{cases} \quad (21)$$

Furthermore, the static moments S_1 , S_2 and S_3 of the three deformation zones relative to the geometric center axis (x-axis) can be calculated respectively through Eq. (22).

$$\begin{cases} S_1 = \int_{C_4}^d 2y\sqrt{d^2 - y^2} dy \\ S_2 = -(S_1 + S_3) \\ S_3 = \int_{-d}^{C_5} 2y\sqrt{d^2 - y^2} dy \end{cases} \quad (22)$$

The moments of inertia I_1 , I_2 and I_3 of the three deformation zones relative to the geometric center layer can be respectively calculated through Eq. (23).

$$\begin{cases} I_1 = \int_{C_4}^d 2y^2\sqrt{d^2 - y^2} dy \\ I_2 = \frac{\pi}{4}(d^4 - c^4) - I_1 - I_3 \\ I_3 = \int_{-d}^{C_5} 2y^2\sqrt{d^2 - y^2} dy \end{cases} \quad (23)$$

Similarly, when the elastic-plastic boundary is located between the geometric center layer and the inner radius, as shown in Fig. 4(b), the areas of the three deformation zones can be calculated as:

$$\begin{cases} A_1 = \int_c^d 2\sqrt{d^2 - y^2} dy + \int_{C_4}^c 2(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}) dy \\ A_2 = \int_{C_5}^{C_4} 2(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}) dy \\ A_3 = \int_{-d}^{-C_5} 2\sqrt{d^2 - y^2} dy + \int_{-C_5}^c 2(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}) dy \end{cases} \quad (24)$$

The static moments of the three deformation zones can be calculated by Eq. (25).

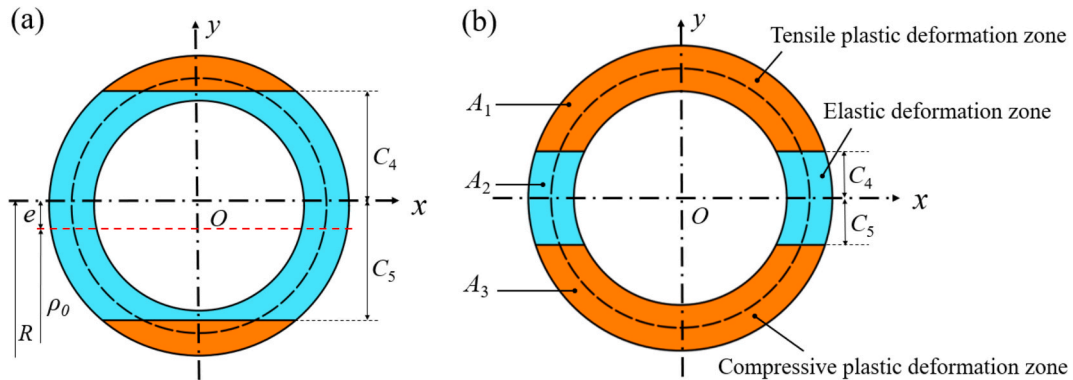


Fig. 4. Elastic-plastic evolution process of the tube bending cross-section: (a) The elastic-plastic boundary is located between the inner radius and the outer radius; (b) The elastic-plastic boundary is located between the inner radius and the geometric center layer (x-axis).

$$\begin{cases} S_1 = \int_c^d 2y\sqrt{d^2 - y^2} dy + \int_{C_4}^c 2y\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy \\ S_2 = \int_{C_5}^{C_4} 2y\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy \\ S_3 = \int_{-d}^{-c} 2y\sqrt{d^2 - y^2} dy + \int_{-c}^{C_5} 2y\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy \end{cases} \quad (25)$$

Further, the moments of inertia corresponding to the three deformation zones can be calculated by Eq. (26).

$$\begin{cases} I_1 = \int_c^d 2y^2\sqrt{d^2 - y^2} dy + \int_{C_4}^c 2y^2\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy \\ I_2 = \int_{C_5}^0 2y^2\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy + \int_0^{C_4} 2y^2\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy \\ I_3 = \int_{-d}^{-c} 2y^2\sqrt{d^2 - y^2} dy + \int_{-c}^{C_5} 2y^2\left(\sqrt{d^2 - y^2} - \sqrt{c^2 - y^2}\right) dy \end{cases} \quad (26)$$

Based on the classical elastic-plastic theory and the principle of axial static equilibrium, substituting the above results into Eq. (27), the elastic-plastic boundaries and the bending radius of neutral layer can be calculated respectively.

$$\begin{cases} F_L = \int_{A_1} \left[K \left(\frac{R - \rho_0 + y}{\rho_0} + b \right)^n \right] dA + \int_{A_2} \left(\frac{R - \rho_0 + y}{\rho_0} \right) E y dA \\ \quad + \int_{A_3} \left[-K \left(-\frac{R - \rho_0 + y}{\rho_0} + b \right)^n \right] dA \\ G_4 = \rho_0 \varepsilon_s - (R - \rho_0) \\ G_5 = -\rho_0 \varepsilon_s - (R - \rho_0) \\ E \varepsilon_s = K(\varepsilon_s + b)^n \end{cases} \quad (27)$$

Finally, the obtained variable results are substituted into Eq. (28), the bending moment can be calculated.

$$\begin{aligned} M = \int_{A_1} \left[K \left(\frac{R - \rho_0 + y}{\rho_0} + b \right)^n \right] y dA + \int_{A_2} \left(\frac{R - \rho_0 + y}{\rho_0} \right) E y dA \\ + \int_{A_3} \left[-K \left(-\frac{R - \rho_0 + y}{\rho_0} + b \right)^n \right] y dA \end{aligned} \quad (28)$$

According to the assumptions, the unloading process is a completely elastic recovery process. The unloading law of classical elastoplastic theory indicates that if the initially loaded object is regarded as an elastic body, when a load of the same magnitude, opposite direction and the same point of application as the original load is applied to it, the loaded object will only undergo elastic deformation, and the deformation is equivalent to the elastic recovery amount during the unloading of the original load. Therefore, the unloading process of bending can be

equivalently regarded as a reverse elastic bending process.

After an opposite bending moment M_e is applied to the formed tube, the elastic strain distribution state generated in the cross-section is shown in Fig. 5(a). The bending radius generated under the bending moment is R' , and the bending radius of the neutral layer is ρ' . When the tube undergoes springback after unloading, the elastic deformation disappears, which causes the bending radius to increase, and the corresponding bending radius is R_f , the bending radius of the neutral layer is ρ_f , as shown in Fig. 5(b).

The residual strain after unloading can be considered as the result of the superposition of the bending strain and the strain during the springback process, which can be expressed as:

$$\varepsilon_f = \varepsilon + \varepsilon' \quad (29)$$

where ε represents the total strain generated during the bending process, ε' is the elastic strain during the equivalent springback process and ε_f denotes the residual strain after springback.

Further, the strains at different stages can be described as:

$$\begin{cases} \varepsilon = \frac{(R - \rho_0) + y}{\rho_0} \\ \varepsilon' = -\frac{(\rho' - R') + y}{\rho'} \\ \varepsilon_f = \frac{(R_f - \rho_f) + y}{\rho_f} \end{cases} \quad (30)$$

Combining Eq. (29) and (30), the springback geometric constraint equation can be obtained.

$$y \left(-\frac{1}{\rho_0} + \frac{1}{\rho'} + \frac{1}{\rho_f} \right) = \left(\frac{R}{\rho_0} + \frac{R'}{\rho'} - \frac{R_f}{\rho_f} \right) - 1, \quad -d \leq y \leq d \quad (31)$$

Since the springback equivalent process is pure bending and only produces elastic strain, the reverse equivalent stress σ' can be calculated by Eq. (32).

$$\sigma' = E \varepsilon' = -\frac{E}{\rho'} (y + \rho' - R') \quad (32)$$

According to the principle of axial force equilibrium, assuming the cross-sectional area of the tube is A , the axial force F_e and the bending moment M_e can be calculated respectively by Eqs. (33) and (34).

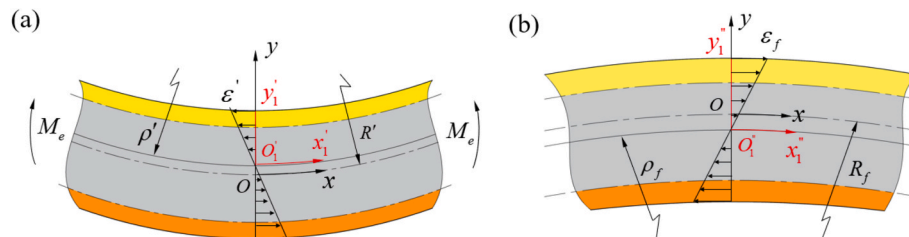


Fig. 5. Strain distribution before and after tube bending unloading: (a) Strain distribution of the equivalent process of unloading springback; (b) Residual strain distribution after unloading.

$$F_e = -\frac{E}{\rho'} \left[\int_A y dA + (\rho' - R') \int_A dA \right] = -\frac{EA}{\rho'} (\rho' - R') = 0 \quad (33)$$

$$M_e = \int_A \sigma' y dA = -\frac{E}{\rho'} \left[\int_A y^2 dA + (\rho' - R') \int_A y dA \right] = -\frac{EI}{\rho'} = -M \quad (34)$$

By solving the above equations simultaneously, the bending radius after springback can be obtained.

$$R_f = \frac{REI}{EI - \rho_0 M} \quad (35)$$

It should be noted that the calculation method is based on the Swift constitutive model and axial force equilibrium equation, incorporating neutral layer shifting to further establish a springback solution framework. Specifically, parameters such as the strength coefficient, strain hardening exponent and yield strength of the material can be obtained through room-temperature tensile tests. By combining the outer and inner radii of the tube, the stress-strain distribution before and after bending can be calculated to determine the springback radius. Therefore, this model is currently primarily applicable to plane bending of isotropic thin-walled metal tubes. For anisotropic materials, the constitutive model needs to be modified within this framework and further validated.

2.4. Springback radius estimation based on WOA-Attention-BP

The free bending springback of tubes is affected by the coupled action of multiple factors such as material properties, geometric parameters, and process parameters, and exhibits strong nonlinear characteristics. Although traditional analytical models and finite element methods can describe the physical process, their computational cost is high, making it difficult to meet the demand for fast and high-precision prediction and impossible to apply to industrial scenarios requiring real-time online prediction and adaptive control. Therefore, this paper constructs a WOA-Attention-BP model for predicting springback radius, and its overall workflow is shown in Fig. 6.

The model first uses the Attention mechanism to adaptively weight the input features, accurately quantifying the contribution of each parameter to the springback behavior and highlighting the features that have a greater influence on the springback radius. It then uses the Whale Optimization Algorithm (WOA) to globally optimize the initial weights and biases of the BP neural network and the parameters related to the attention mechanism, so as to reduce the sensitivity of the network to initial parameters and improve the prediction accuracy and convergence performance of the model.

2.4.1. WOA optimization algorithm

The Whale Optimization Algorithm (WOA) is a swarm-intelligence optimization algorithm that simulates the predatory behavior of whales. It mainly includes a three-stage predation mechanism of “surrounding the prey-spiral approaching-global searching,” mapping the optimization problem to the prey-hunting process, and it has the

advantages such as simple structure, few hyperparameters and strong convergence [30,31]. The parameter optimization mechanism and the core iterative process of the algorithm are as follows:

First, Whales contract the encirclement by tracking the current optimal individual position, and the specific expressions are shown in Eqs. (36)–(38).

$$\vec{\theta}(t+1) = \vec{\theta}'(t) - \vec{\lambda} \cdot \left| \vec{\mu} \cdot \vec{X}'(t) - \vec{X}(t) \right| \quad (36)$$

$$\vec{\lambda} = 2 \vec{a} \cdot \vec{r} - \vec{a} \quad (37)$$

$$\vec{\mu} = 2 \cdot \vec{r} \quad (38)$$

where t is the current number of iterations, $\vec{\theta}(t)$ is the position vector of an individual whale, $\vec{\theta}'(t)$ is the position vector of the current target prey, $\vec{r}$ is a random vector within the interval $[0, 1]$, and $\vec{a}$ is a coefficient vector. The coefficient vectors $\vec{\lambda}$ and $\vec{\mu}$ dynamically adjust the search range, guiding the population to converge toward the global optimal region.

Secondly, during the local exploitation stage, the whale approaches the prey in a spiral path, simulating the fine behavior of the bubble-net contraction for hunting. This process is cooperatively achieved through two mathematical description methods, the shrinking encircling mechanism and the spiral updating mechanism, aiming to enhance the refined search capability of the algorithm in the vicinity of the local optimal solution.

The shrinking encircling mechanism achieves the gradual contraction of the search range by dynamically adjusting the convergence vector $\vec{a}$. This coefficient linearly decreases from 2 to 0 during the iteration process, as shown in Eq. (39).

$$\vec{a} = 2 \cdot \left(1 - \frac{t}{t_{\max}} \right) \quad (39)$$

The spiral updating mechanism first determines the spatial distance between the whale's position $\vec{\theta}(t)$ and the target prey $\vec{\theta}'(t)$, and then construct a logarithmic spiral movement trajectory equation based on this distance to achieve the spiral update of the whale's position, which can be expressed as:

$$\vec{\theta}(t+1) = \left| \vec{\theta}'(t) - \vec{\theta}(t) \right| \cdot e^{g\xi} \cdot \cos(2\pi\xi) + \vec{\theta}'(t) \quad (40)$$

where g is used to control the geometric shape of the logarithmic spiral curve, and ξ is a random number between $[-1, 1]$.

To simultaneously achieve shrinking encircling and spiral update, the algorithm dynamically selects the update strategy through a random probability $p \in [0, 1]$, as shown in Eq. (41).

$$\vec{\theta}(t+1) = \begin{cases} \vec{\theta}'(t) - \vec{\lambda} \cdot \left| \vec{\mu} \cdot \vec{\theta}'(t) - \vec{\theta}(t) \right|, & p < 0.5 \\ \left| \vec{\theta}'(t) - \vec{\theta}(t) \right| \cdot e^{g\xi} \cdot \cos(2\pi\xi) + \vec{\theta}'(t), & p \geq 0.5 \end{cases} \quad (41)$$

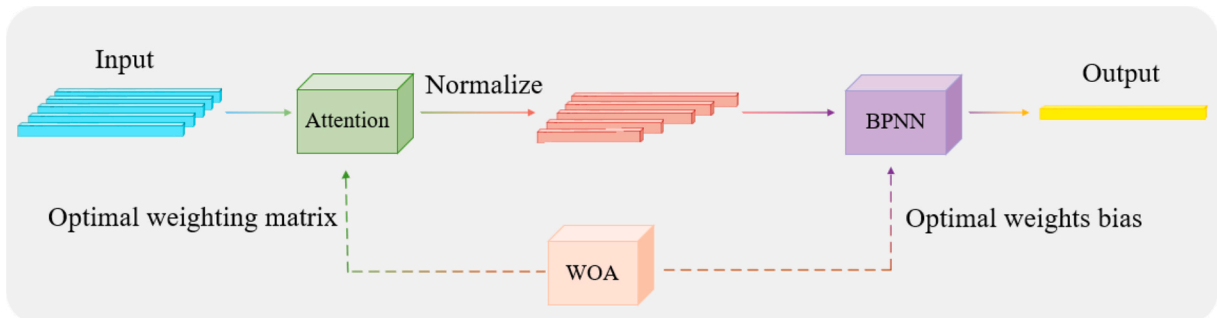


Fig. 6. Training flowchart of the WOA-Attention-BP model.

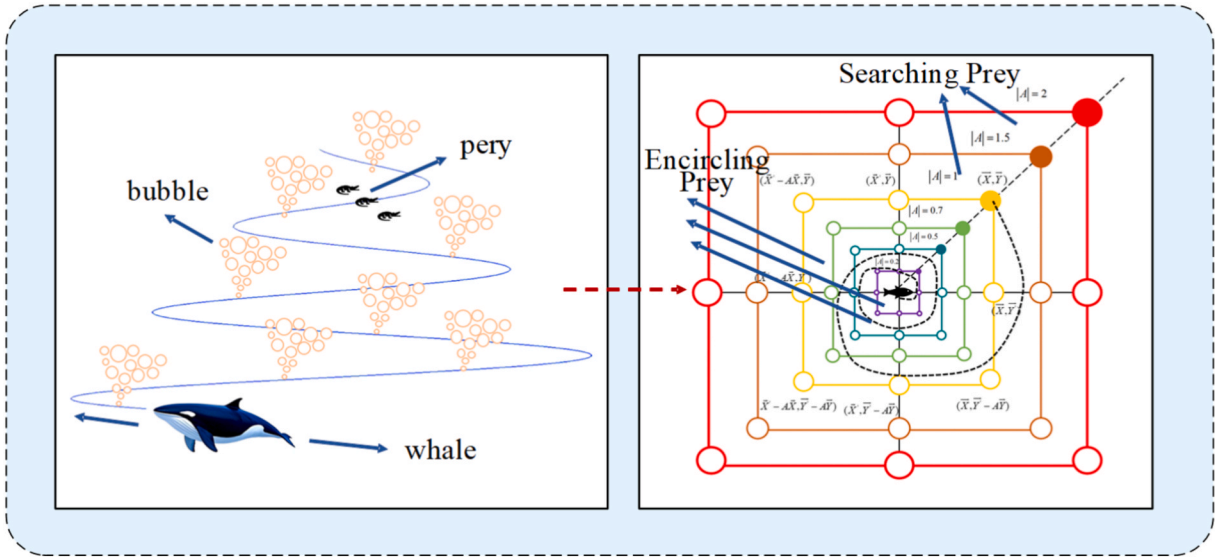


Fig. 7. Schematic diagram of the bionic predatory and search mechanisms in the Whale Optimization Algorithm (WOA).

Finally, when the algorithm fails to obtain valid prey information during the iterative process, the whale will abandon the current local exploitation strategy and switch to a global random exploration stage. This mechanism aims to prevent the algorithm from getting stuck in a local optimum and enhances its global search capability in the solution space. Specifically, when $|\vec{\lambda}| > 1$, the whale randomly selects individuals for global exploration and updates their positions based on this, which can be described as:

$$\vec{\theta}(t+1) = \vec{\theta}_{rand}(t) - \vec{\lambda} \cdot \left| \vec{\mu} \cdot \vec{\theta}_{rand}(t) - \vec{\theta}(t) \right| \quad (42)$$

where $\vec{\theta}_{rand}$ is the random position vector of a whale.

As shown in Fig. 7, the WOA algorithm dynamically regulates the predatory behavior of whales through the value of the coefficient vector $|\vec{\lambda}|$, determining whether the whales enter the search-for-prey stage or the encircling-prey stage, thereby achieving an intelligent switch between the two modes of “global exploration” and “local exploitation.” When $|\vec{\lambda}| \geq 1$, the algorithm determines that there is currently a lack of effective prey location information, and the whales enter the search-for-prey stage. Conversely, when $|\vec{\lambda}| < 1$, the algorithm determines that a potential optimal solution region has been detected, and the whales enter the encircling-prey stage. At this point, the algorithm implements shrinking encircling and spiral updating strategies around the current optimal solution (prey location), focusing on the refined exploitation of local areas to improve convergence accuracy.

2.4.2. Attention mechanism

The core idea of the attention mechanism is to allow the model to adaptively learn the importance distribution of input features during the training process, thereby enabling the model to focus on key information [39]. Therefore, to improve the model's ability to identify the importance of different input features, an attention mechanism is introduced before the BP neural network to adaptively weight each input feature. The specific process is as follows:

First, as shown in Eq. (43), the original five-dimensional input feature vector is linearly transformed through a learnable weight matrix N and projected into an n -dimensional hidden feature space. Subsequently, the hyperbolic tangent function is utilized for transformation to enhance the nonlinear representation capability of the features.

$$\mathbf{h} = \tanh(N \cdot \mathcal{D}) \quad (43)$$

Then, a trainable fully connected weight vector W is used to perform a linear transformation on the hidden feature vector $\mathbf{h}$ to obtain the attention score ζ for each feature, which preliminarily characterizes the contribution of each input feature to the springback.

$$\zeta = W \cdot \mathbf{h} \quad (44)$$

Subsequently, the Softmax function is applied for normalization to generate the final normalized attention weight vector α , as shown in Eq. (45).

$$\alpha_i = \frac{e^{\zeta_i}}{\sum_{j=1}^{N_f} e^{\zeta_j}} \quad (45)$$

where ζ_i denotes the raw attention score of the i -th feature, and α_i is the corresponding normalized weight.

Finally, as shown in Eq. (46), the normalized attention weight vector is multiplied element-wise with the original input feature vector to generate the final output feature vector weighted by the Attention mechanism.

$$\alpha'_i = \alpha_i \mathcal{D}_i \quad (46)$$

The specific principle and process are shown in Fig. 8. Before the features are input into the BP neural network, the attention mechanism can automatically learn the importance of each feature to the springback, and assign different weights to different features without the need to manually set the feature weights. Therefore, the attention mechanism can enhance the model's learning ability for key features, while suppressing the focus on irrelevant features, thereby improving the model's recognition accuracy of important features and intuitively reflecting the importance of the features.

2.4.3. BP neural network

The Back Propagation Neural Network (BPNN) is a multi-layer feedforward neural network that is trained through the error back-propagation algorithm, thereby achieving a nonlinear mapping relationship between input and output. As shown in Fig. 9, the BPNN constructed in this paper adopts a classic three-layer structure, consisting of an input layer, a hidden layer, and an output layer. Information is transmitted forward and errors are corrected backward through weighted connections among neurons in each layer. The input layer is composed of five neurons, namely: bending die offset, yield strength,

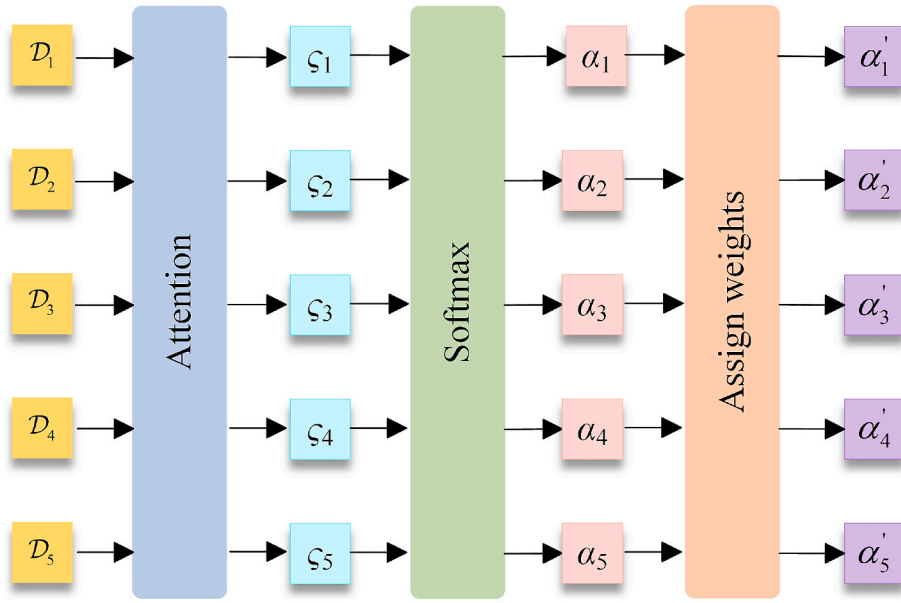


Fig. 8. Schematic diagram of the weighted attention mechanism calculation principle.

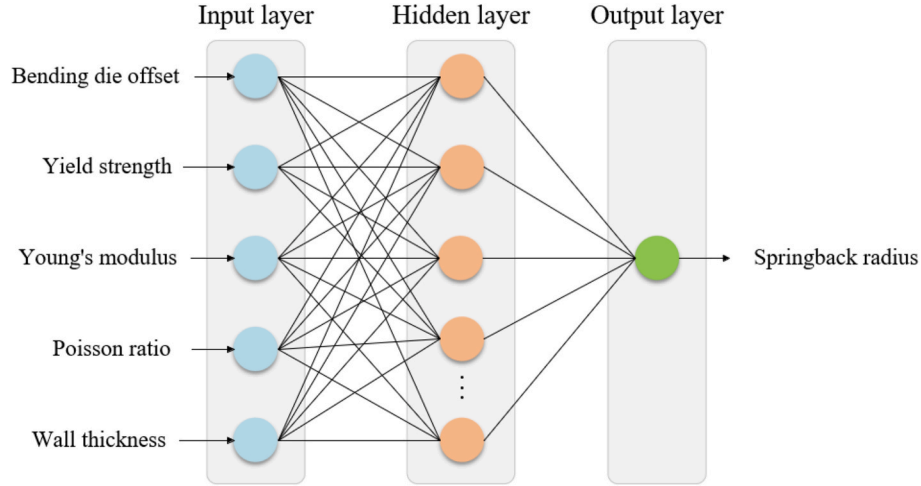


Fig. 9. Structure diagram of the BP neural network for tube bending springback prediction.

Young's modulus, Poisson's ratio, and wall thickness. The output layer is a single-neuron structure corresponding to the springback radius of the tube. The specific calculation process is as follows:

First, the connection weights $w_{ij}^{(\ell)}$ and biases $\beta_j^{(\ell)}$ between layers in the network are randomly initialized, the weights are used for signal transmission between different neurons, and biases are used to adjust the activation threshold of neurons.

Then, for the input data samples, the neural network calculates the output values of each neuron through a hierarchical progressive approach. The core of this process lies in performing a weighted sum operation on the input of each neuron and using an activation function to achieve a nonlinear mapping, thereby completing the layer-by-layer transmission of information, as shown in Eq. (47).

$$o_j^{(\ell)} = f \left(\sum_{i=1}^{N_{\ell-1}} w_{ij}^{(\ell)} o_i^{(\ell-1)} + \beta_j^{(\ell)} \right) \quad (47)$$

where $w_{ij}^{(\ell)}$ is the weight connecting the i -th neuron in the $(\ell-1)$ -th layer to the j -th neuron in the ℓ -th layer, $\beta_j^{(\ell)}$ is the bias of the j -th neuron in the ℓ -th layer, and $o_j^{(\ell)}$ is the output of the j -th neuron in the ℓ -th layer.

Further, the predicted values of the network's output layer are compared with the actual target values to calculate the error between the predicted and actual values. Based on the output error, the error signal of the output layer is backpropagated along the network layers in reverse using the chain rule of differentiation to calculate the contribution of each neuron to the total error. Subsequently, the gradient descent method is employed to adjust the network weights and bias parameters according to the preset learning rate, as shown in Eqs. (48) and (49). This mechanism continuously reduces the network error through iterative optimization, thereby gradually improving the model's fitting performance.

$$w_{ij}^{(\ell)} = w_{ij}^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} \quad (48)$$

$$\beta_j^{(\ell)} = \beta_j^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial \beta_j^{(\ell)}} \quad (49)$$

where $\mathcal{L}$ is the loss function value of the output layer, and η is the learning rate.

Finally, the complete process of forward propagation, error calculation, backpropagation, and parameter update are executed in a loop

until the model output meets the preset accuracy requirements or the termination conditions are reached. Through this iterative optimization mechanism, the neural network gradually corrects the parameter configuration and continuously enhances its feature extraction ability and prediction performance for the input data.

3. Results and discussion

3.1. Experimental evaluation

To verify the accuracy of the springback radius measurement method described in Section 2.2, bending experiments were conducted on $20 \times 1 \text{ mm}^2$ (diameter $\times$ wall thickness) Al6061 aluminum alloy tubes by using the three-axis free bending device, as shown in Fig. 10(a). For the thin-walled tube, when $R/D < 4$ (bending radius to outer diameter ratio), wrinkling and cross-sectional collapse become the primary failure modes, rendering the model's assumptions invalid. When $R/D > 10$, the engineering significance of springback prediction diminishes. Therefore, bending experiments were conducted within the range ($R/D = 5\text{--}8$) where stable forming and significant springback behavior occur. After the tube was bent and unloaded at room temperature (25°C), the coordinates of the central axis in the constant curvature section were fitted to further obtain the springback radius. Specifically, the 3D laser scanner, FreeScan UE11, was used to perform full-scale digital scanning of the bent component to obtain actual bending profiles and surface topography data for subsequent processing. The scanner has a scanning accuracy of $\pm 0.02 \text{ mm}$, a maximum scanning speed of 1,350,000 points per second, and a scanning range covering 0.05 mm to 3000 mm. Before starting the measurement, the scanner must be calibrated using the provided calibration board. Fig. 10(c) shows the complete springback radius measurement chain, including scanner calibration, workpiece placement, multi-angle scanning, point cloud registration, outlier removal, surface model reconstruction central axis extraction, constant curvature segment selection, coordinate sampling and circle fitting.

To quantitatively evaluate the measurement uncertainty of the proposed springback radius measurement method, the influence of coordinate measurement errors on the fitted radius was first investigated. According to Section 2.2.2, the propagation of coordinate measurement errors into radius uncertainty is highly dependent on the fitted arc angle ϕ . Taking the nominal bending radius of $R = 120 \text{ mm}$ as an example, the propagation characteristics of coordinate noise into radius errors under different fitted arc angles after springback were analyzed, and the results are presented in Table 1. As shown in Fig. 11(a), the amplification factor decreases

significantly with increasing fitted arc angle and gradually reaches a stable value, indicating that circle fitting under short-arc conditions is more sensitive to perturbations in the extracted centerline coordinates.

Furthermore, a fitted arc angle of $\phi = 60^\circ$ was selected as an example, and the propagation of coordinate measurement uncertainty into the fitted radius was statistically verified using the Monte Carlo method, as shown in Fig. 11(b). The results indicate that the fitted radius approximately follows a normal distribution, with a mean value of $\mu_{R_m} = 134.72 \text{ mm}$ and a standard deviation of $\sigma_R = 0.21 \text{ mm}$. This demonstrates that coordinate measurement errors mainly introduce random fluctuations of the fitted radius around its mean value, without causing significant systematic bias. Therefore, the effectiveness of the proposed uncertainty evaluation method is further verified.

Since multiple error sources are involved in practical radius measurement, the combined standard uncertainty u_C consists of the Type A uncertainty component obtained from repeated scanning experiments and Type B uncertainty components, including scanner calibration error, centerline extraction error, circle fitting residual, and sampling interval error. Furthermore, the tube specimen was independently scanned three times under identical measurement conditions. Each measurement involved repositioning, re-scanning, extracting centerline, and fitting circle. The obtained uncertainty budget is summarized in Table 2. According to Eq. (17), the expanded uncertainty is 0.86 mm.

To further verify the reliability of the proposed springback radius measurement method, its uncertainty and repeatability were compared with those obtained using conventional manual radius gauge and vernier caliper measurements. The radius gauge determines the radius by contact matching between standard circular templates and the curved profile of the specimen, whereas the vernier caliper method calculates the radius based on the projected geometry of the bent specimen onto a plane. As shown in Fig. 11(c), the proposed method achieved an expanded uncertainty of 0.86 mm and a repeatability standard deviation (SD) of 0.63 mm, which were lower than those obtained by the manual radius gauge (1.53 mm and 1.28 mm) and the vernier caliper (2.47 mm and 1.85 mm). These results indicate that conventional measurement methods are affected by operator alignment errors and limited instrument resolution, resulting in increased measurement uncertainty. In contrast, the proposed method integrates complete profile acquisition and automated geometric fitting, improving the accuracy and repeatability of springback radius measurement.

The springback radii of bent tubes under different bending radii were measured following the above procedure. The springback radii obtained by fitting the experimental measurements were compared with the

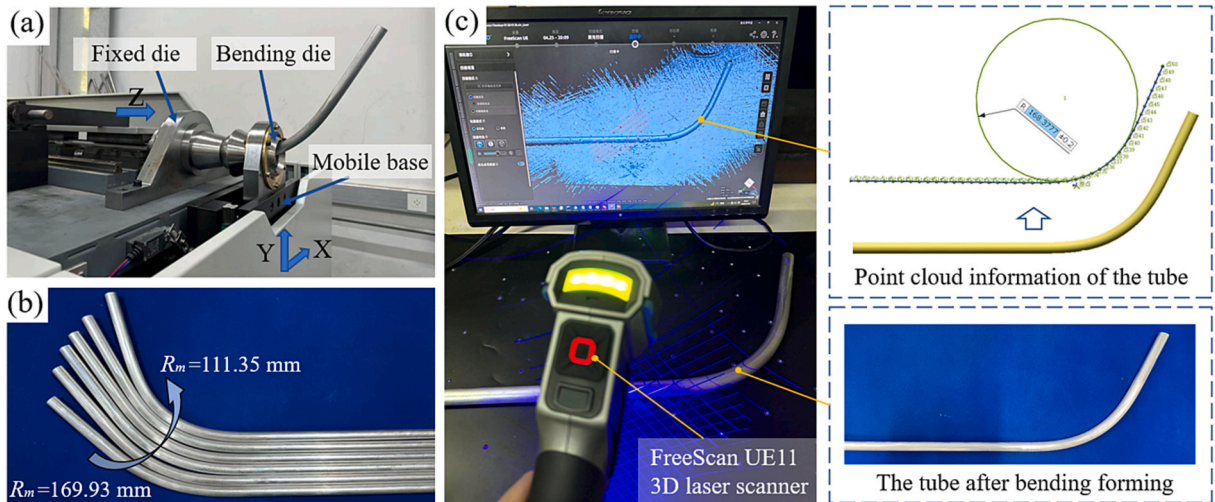
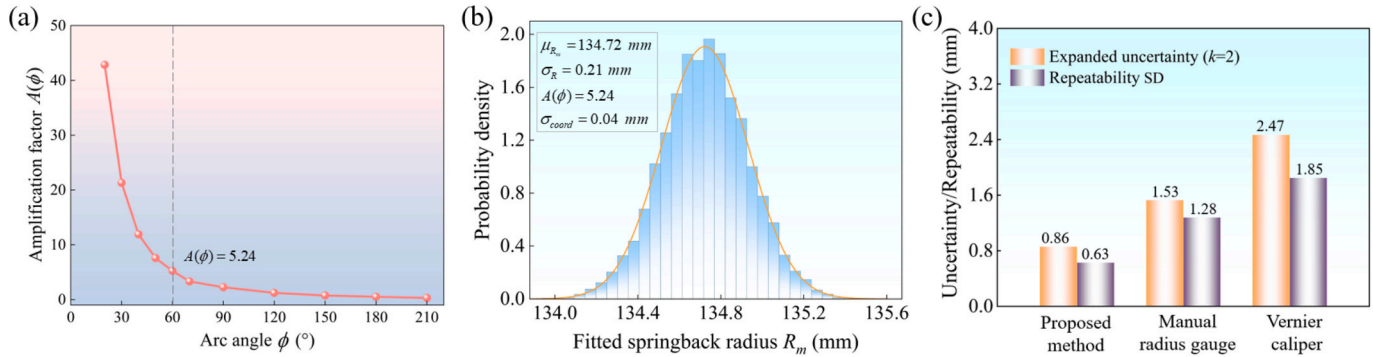


Fig. 10. Experiment and scanning process of tube free bending forming: (a) Tube free bending forming device; (b) Bending components after bending forming; (c) Scanning to extract the bending radius of the bent component.

Table 1

The relationship between the magnification factor of arc fitting and the arc angle.

Arc angle $\phi(^{\circ})$	20	30	40	50	60	70	90	120	150	180	210
Amplification factor $A(\phi)$	42.85	21.3	11.9	7.58	5.24	3.31	2.25	1.22	0.74	0.51	0.33

**Fig. 11.** Uncertainty analysis and measurement performance evaluation of the proposed springback radius measurement method: (a) Influence of fitted arc angle on radius error amplification factor; (b) Monte Carlo simulation of fitted springback radius uncertainty; (c) Comparison of uncertainty and repeatability among different measurement methods.**Table 2**

Measurement uncertainty budget for the springback radius (Nominal bending radius is 120 mm).

Uncertainty source	Type	Estimate (mm)	u_i (mm)
Repeatability of three repeated scanning	A	0.63	0.364
Scanner calibration error	B	0.02	0.012
Centerline extraction error	B	0.017	0.01
Circle-fitting residual	B	0.21	0.21
Sampling interval error	B	0.16	0.092
Combined standard uncertainty u_C	—	—	0.43
Expanded uncertainty U_F ($k=2$)	—	—	0.86

estimates from the analytical model, providing a reference for evaluating the proposed measurement method. As shown in Table 3, the repeatability standard deviation for the measurements across all groups ranges from 0.63 to 0.93 mm, indicating that the established measurement chain has good repeatability. The absolute deviation between the analytical model predictions and experimental measurement means ranges from 6.72 to 11.56 mm. Additionally, the relative deviations range from 6.03% to 6.80%.

In addition, to more clearly illustrate the impact of bending radius variation on the model's prediction accuracy, the data were plotted in Fig. 12. Fig. 12(a) shows the variation of predicted values from the analytical model and experimental measurements with respect to the nominal bending radius. Fig. 12(b) presents the relative deviations at different nominal bending radii. The deviation between the model predictions and experimental measurements gradually increases as the nominal bending radius increases. This indicates that the established analytical model can stably describe the overall trend of springback

radius variation with nominal bending radius. However, owing to factors such as material constitutive, contact friction, cross-sectional deformation and simplifications in the unloading process, the predicted results still exhibit certain systematic deviations.

3.2. Finite element simulation validation

Since the analytical model is based on certain assumptions, there exists a systematic deviation between the predicted results and experimental measurements. To further validate the accuracy of the proposed springback radius measurement method, the measured radii are indirectly assessed by comparing them with independent finite element reference values. Therefore, this section simulates the free bending and springback of tubes. Meanwhile, the element simulation model is further employed to generate training datasets.

3.2.1. Establishment of FE model

The Al6061 aluminum alloy tubes used in this study were produced via hot extrusion, with their specific chemical composition shown in Table 4. Standard tensile specimens were prepared along the axial direction of the tube (20x1 mm²) using the wire cutting machine. The tensile specimens were clamped in a universal tensile testing machine's dedicated fixture, three repeated experiments were conducted at the strain rate of 0.001 s⁻¹, and the average of the three sets of tensile data was used for analysis. As shown in Fig. 13(b), the Swift hardening model was used to fit the stress-strain curve, with a fitting accuracy of 0.996. The obtained mechanical properties are shown in Table 5.

The key components of the design model of the free bending forming device were imported into the ABAQUS/Explicit software. As shown in Fig. 13(a), the free bending FE model includes spherical bearing, bending

Table 3

Comparison between analytical model predictions and 3D scanning measurement results at different nominal bending radii.

Sample number	Nominal bending radius R (mm)	Analytical model prediction R_a (mm)	Mean of three measurements $\bar{R}_m$ (mm)	Repeatability standard deviation S_R (mm)	Absolute deviation R_d (mm)	Relative deviation δ (%)
1	100	104.63	111.35	0.65	6.72	6.03
2	110	115.15	122.82	0.80	7.67	6.25
3	120	125.76	134.54	0.63	8.78	6.53
4	130	136.28	145.86	0.92	9.58	6.57
5	140	147.11	157.62	0.75	10.51	6.67
6	150	158.37	169.93	0.93	11.56	6.80

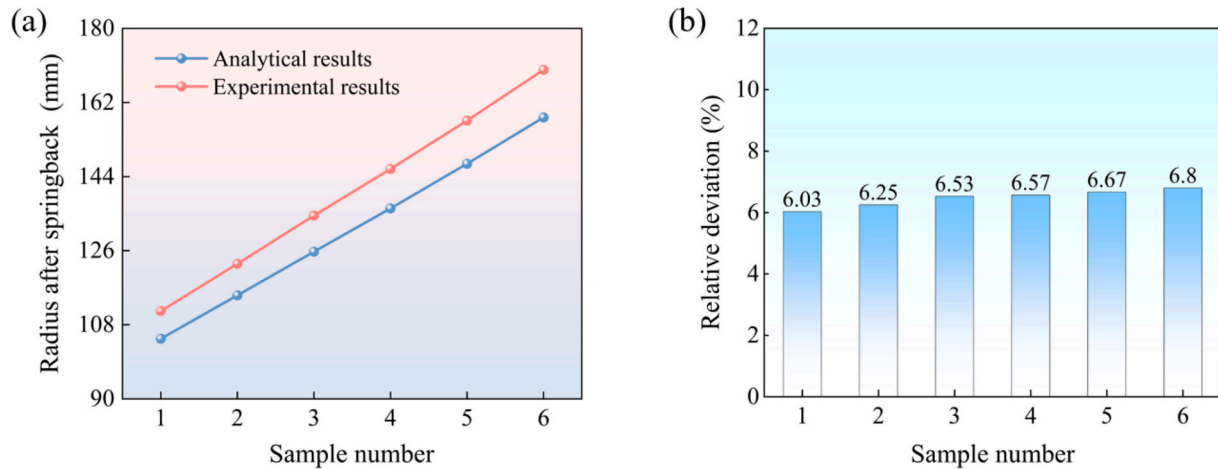


Fig. 12. Predicted results of the bending radius after springback and prediction deviations of the analytical model: (a) Comparison between predicted results of the analytical model and experimental results; (b) Relative deviations of the prediction results of the analytical model.

Table 4

Chemical composition of Al6061 aluminum alloy (mass fraction, %).

Mg	Si	Fe	Cu	Zn	Mn	Cr	Ti	Al
0.58	0.43	0.13	0.03	0.01	0.01	0.02	0.01	Bal.

die, fixed die and thin-walled tube. Among them, the tube was defined as a deformable body and meshed with four-node linear shell elements (S4R), whose mesh size was 1 mm and four layers of mesh along the thickness direction of the tube. The spherical bearing, bending die and fixed die were all defined as rigid bodies, whose mesh types were R3D4. To improve the calculation efficiency, a coarser mesh was adopted for non-critical components, whose mesh size was set to 3. In addition, the penalty function was used to define the contact between different components, with the sliding mode being finite sliding and the normal contact behavior being hard contact. Considering the application of lubricant between the tube and the

dies during the actual bending process, and referring to previous research [40–42], the friction coefficient between the tube and the bending die surface was set to 0.04, while the friction coefficient between other components was set to 0.06. Throughout the simulation process, a fixed constraint was applied to the fixed die, a displacement was applied to the spherical bearing, and the rear end face of the thin-walled tube was coupled with a reference point. The feed of the tube was achieved by setting the forward speed of the reference point. To reduce computation time, the mass scaling factor was set to 100. The artificial strain energy (ALLAE), internal energy (ALLIE) and kinematic energy (ALLKE) were extracted during the simulation process for calculation. As shown in Fig. 14(a), both the ratios of ALLKE/ALLIE and ALLAE/ALLIE were less than 5% after load application, which indicated that the quasi-static simulation conditions were satisfied and the simulation results were highly reliable [43,44].

During the explicit dynamic forming process, as the tube exits the bending die, the local contact conditions change and partial elastic recovery occurs. However, at this stage, the feeding boundary conditions

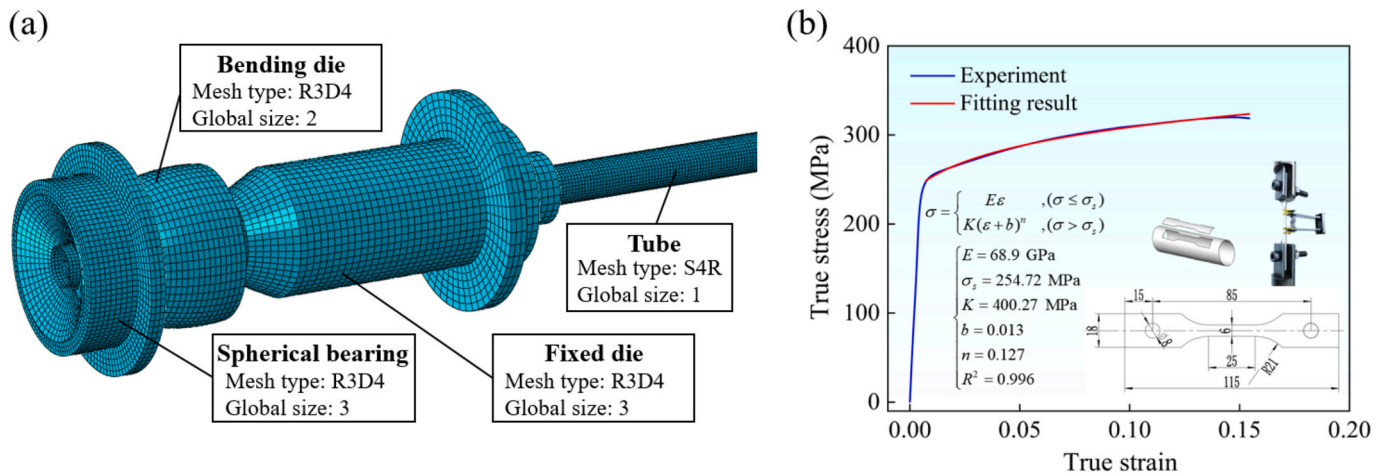


Fig. 13. Finite element simulation model and mechanical property parameter fitting: (a) FE model for tube free bending; (b) Stress-strain curve and fitted parameters.

Table 5

Material property parameters of Al6061 tubes.

Al6061	Material property parameters				ν	n	b
	$\rho / (\text{Kg}/\text{m}^3)$	E / GPa	σ_s / MPa	K / MPa			
ϕ 20x1 mm	2.7×10^3	68.9	254.72	400.27	0.33	0.127	0.013

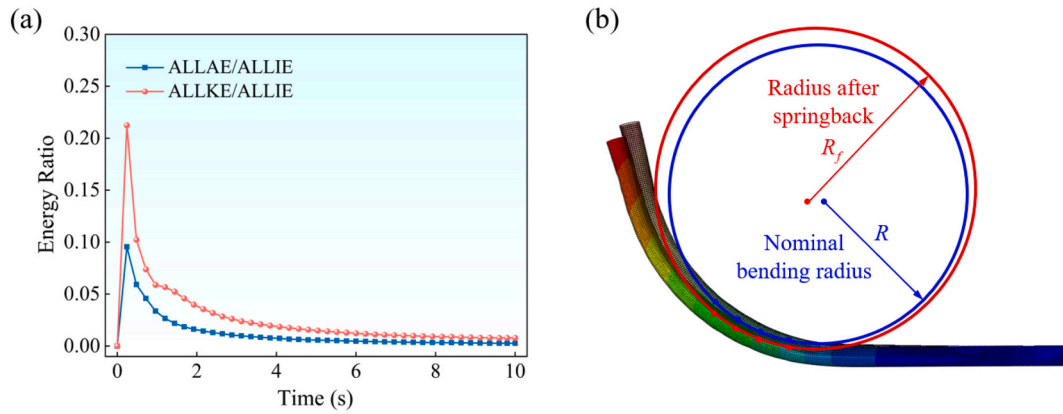


Fig. 14. Simulation of free bending and springback of the tube: (a) Energy changes during the bending forming process; (b) The measurement method of the radius after springback.

and mold contact have not yet been fully released. Thus, the explicit analysis does not represent the final static equilibrium state after complete unloading. To obtain the final springback of the tube with all external loads and contact constraints removed, the stress–strain field at the end of the forming process was taken as the initial condition and imported into the ABAQUS/Standard implicit solver. The quasi-static unloading and springback analysis was then carried out using the Static/General analysis step. Specifically, the elastic recovery process of the tube was achieved by removing the external load. After completing the springback calculation, coordinates of several discrete characteristic points on the stable section of the tube's central layer were extracted, and curve fitting was performed using MATLAB to obtain the bending radius after springback, as shown in Fig. 14(b).

3.2.2. Validation of measurement accuracy by simulation

To verify the accuracy of the proposed springback radius measurement method, the simulated springback results under identical conditions were compared with the springback radii obtained from experimental measurements. Specifically, bending tests were conducted using the free bending forming device under various eccentricities to establish the relationship between the bending radius and eccentricity, as shown in Fig. 15. Based on the obtained functional relationship, the FE simulations were carried out using eccentricities yielding good forming quality, with the specific experimental design shown in Table 6, while the simulation and experimental results are shown in Fig. 16.

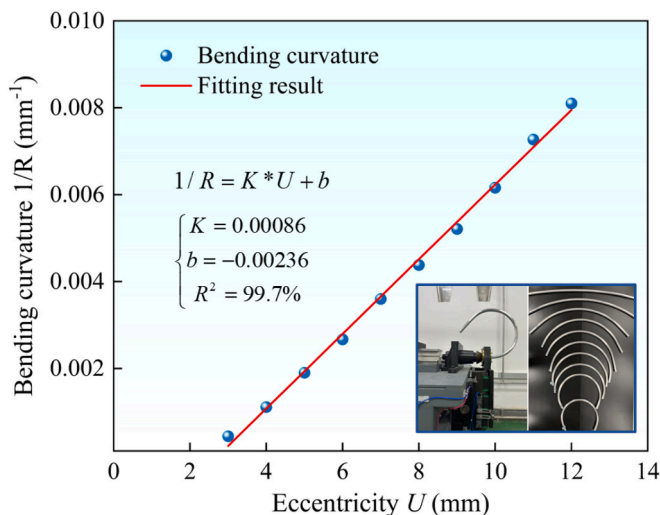


Fig. 15. Relationship between eccentricity and bending radius.

Table 6

Experimental scheme for feasibility validation of FE simulation.

Experimental specimen	Wall thickness (mm)	Bending die offset U (mm)	Feed rate (mm·s ⁻¹)
1	1	12.5	30
2	1	12	30
3	1	11.5	30
4	1	11	30
5	1	10.5	30
6	1	10	30
7	1.5	11	30
8	2	11	30
9	2.5	11	30
10	3	11	30

Fig. 17(a) shows the comparison between the FE predicted springback radius and experimental measurements. It can be clearly observed that the simulation results are generally consistent with the experimental data, showing a high degree of agreement. As shown in the error analysis of Fig. 17(b), within the studied range, the average relative deviation between simulation and experiment is 2.57%, with a maximum relative deviation of 2.91%, indicating that the measurement method has high accuracy.

In addition, to further validate the applicability of the FE model under different wall thickness conditions, additional bending experiments and corresponding simulations were conducted for Al6061 tubes with wall thicknesses ranging from 1.5 to 3.0 mm. As shown in Fig. 17(c), as the wall thickness increases, the relative deviation between simulation predictions and experimental measurements generally tends to increase. When the wall thickness reaches 3.0 mm, the relative deviation reaches 3.67%. These results indicate that the established FE model maintains good predictive consistency within the wall thickness range investigated in this study. However, the model exhibits some dependence on wall thickness, with higher prediction accuracy achieved for thinner-walled tubes ($D/t > 10$).

3.2.3. Analysis of parameters influencing springback radius

The mechanical properties of materials are key factors influencing the deformation behavior of tubes during bending. During the unloading and springback process after free bending, the yield strength plays a decisive role in the stress release and elastic recovery within the material. When the yield strength is high, the critical stress required to cause plastic deformation in the tube increases accordingly. Therefore, during the unloading stage, the tube releases more elastic strain energy internally, resulting in a more significant springback effect. To quantitatively reveal the influence of yield strength on springback, FE simulation analyses were conducted using yield strengths ranging from 180 to

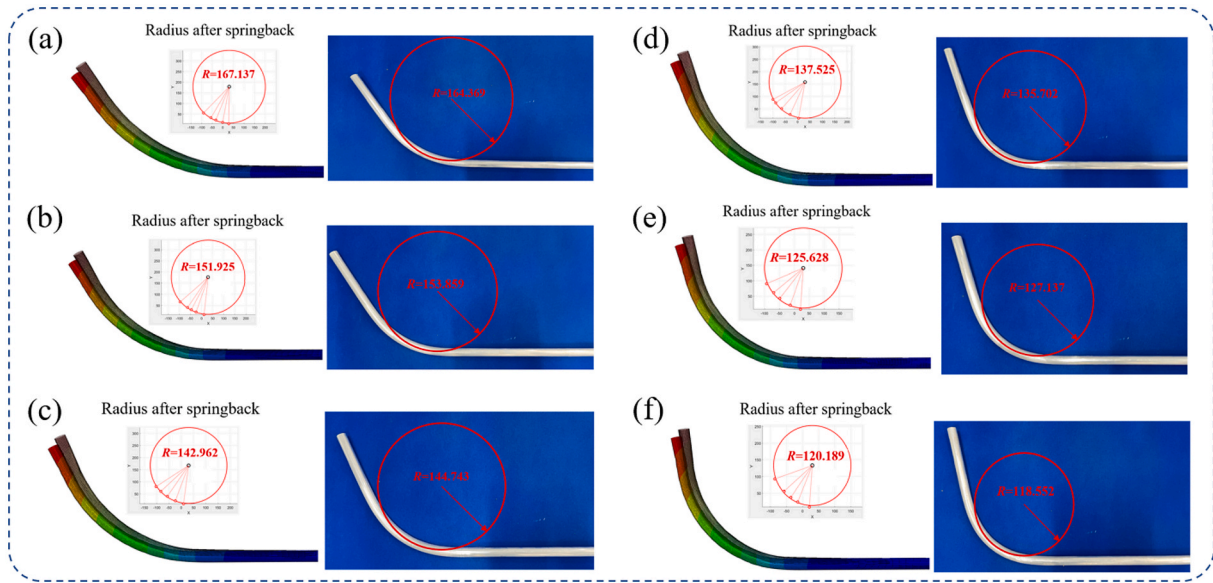


Fig. 16. Comparison between FE simulation and Experimental measurement results under different eccentricities: (a) $U = 10$ mm; (b) $U = 10.5$ mm; (c) $U = 11$ mm; (d) $U = 11.5$ mm; (e) $U = 12$ mm; (f) $U = 12.5$ mm.

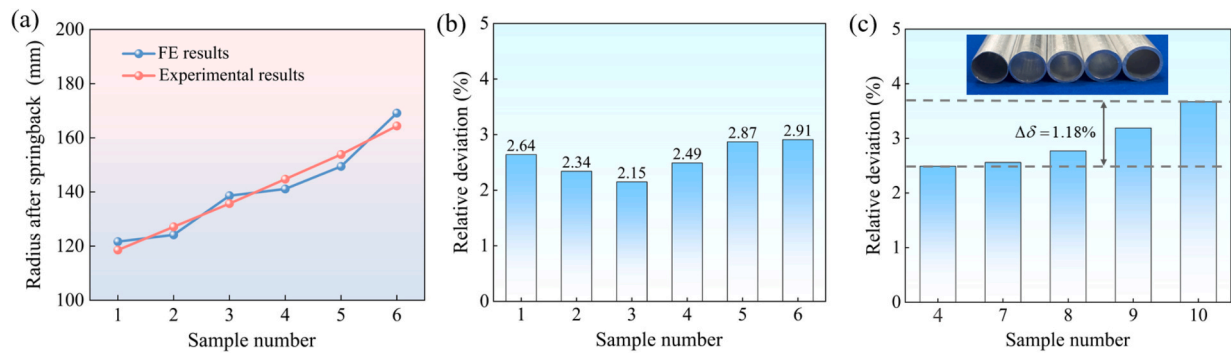


Fig. 17. Comparison of FE simulation results with experimental measurement results and relative deviation: (a) Comparison of FE predicted and experimentally measured springback radii under different eccentricities; (b) Relative deviation between experiment and simulation with same wall thicknesses; (c) Relative deviation between experiment and simulation with different wall thicknesses.

340 MPa as the variable. This range covers the yield levels of Al6061 aluminum alloy under different heat treatment conditions, spanning from the solution-treated and naturally aged state ($T4$, $\sigma_s \approx 180$ MPa) [45] to the artificially aged state ($T6$, $\sigma_s \approx 260$ MPa) [46], and further to the cold-worked strengthened state ($\sigma_s \approx 340$ MPa). To minimize deviations caused by the physical correlation between yield strength and hardening parameters, the hardening exponent n was set to 0.15. The strength coefficient K for each yield strength was then determined in reverse using the Swift hardening model, followed by simulation studies. The specific simulation results are shown in Fig. 18.

In actual tube production, tubes from different batches may exhibit variations in mechanical properties. In addition to yield strength, other material parameters, cross-sectional dimensions and process conditions also affect the springback. Therefore, using the same single-factor control method, the Young's modulus, Poisson's ratio, tube feed rate and wall thickness were further selected as variables for simulation analysis. Specifically, the Young's modulus ($E = 62\text{--}70$ GPa), Poisson's ratio ($\nu = 0.31\text{--}0.35$), tube feed rate ($V = 20\text{--}60$ mm·s⁻¹), and tube wall thickness ($T = 1\text{--}3$ mm) were selected as parameter ranges to simulate and analyze the free bending springback process. Using the method shown in Fig. 14(b), the springback results were extracted, and the results under different parameter conditions were summarized in Table 7.

Fig. 19(a) shows the variation of springback with respect to each

influencing parameter. It can be observed that yield strength and wall thickness exhibit obvious positive correlations with springback. As both yield strength and wall thickness increase, the springback amount also increases gradually. Young's modulus shows a negative correlation trend with springback. This is because materials with higher modulus store less elastic strain energy after forming, resulting in smaller elastic recovery after unloading. The effects of Poisson's ratio and feed rate on springback display fluctuating characteristics. To quantitatively evaluate the influence degree of each parameter on springback, a single-factor variance analysis method was employed for sensitivity analysis. The influence weights of each parameter were calculated using Eqs. (50)–(52), and the results are shown in Table 8. As illustrated in Fig. 19 (b), yield strength has the highest influence weight of 64.82%, making it the key parameter affecting springback. Wall thickness follows next at 21.37%, elastic modulus and feed rate are moderately sensitive parameters with weights of 7.26% and 6.07%, respectively. Poisson's ratio has the weakest impact, accounting for only 0.47%.

For the i -th influencing parameter, the mean springback amount over the five levels can be calculated as

$$\overline{\Delta R}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \Delta R_{ij} \quad (50)$$

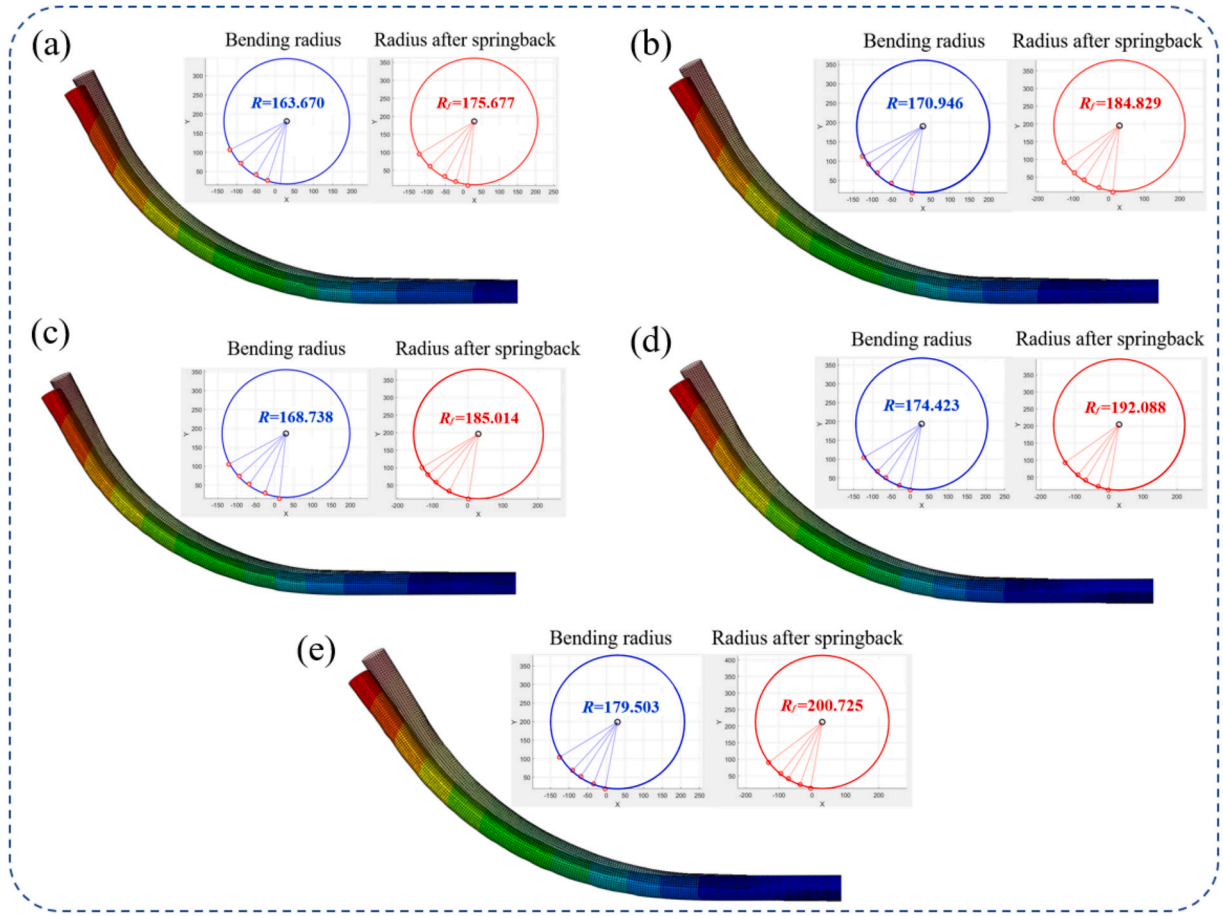


Fig. 18. Springback phenomenon of tube free bending under different yield strengths: (a) $\sigma_s = 180$ MPa; (b) $\sigma_s = 220$ MPa; (c) $\sigma_s = 260$ MPa; (d) $\sigma_s = 300$ MPa; (e) $\sigma_s = 340$ MPa.

Table 7

FE simulation results of bending and springback radii under different influencing parameters.

Influencing parameters	Parameter value	Bending radius R (mm)	Springback radius R_f (mm)	Springback amount ΔR (mm)
σ_s (MPa)	180	163.670	175.677	12.007
	220	170.946	184.829	13.883
	260	168.738	185.014	16.276
	300	174.423	192.088	17.665
	340	179.503	200.725	21.222
E (GPa)	62	139.524	152.893	13.369
	64	139.583	152.174	12.590
	66	142.858	154.419	11.561
	68	144.047	155.004	10.954
	70	141.323	152.787	10.464
ν	0.31	143.390	157.253	13.862
	0.32	142.875	156.776	13.901
	0.33	141.944	157.492	14.548
	0.34	147.097	160.904	13.807
	0.35	144.127	158.217	14.089
V (mm·s ⁻¹)	20	118.259	134.385	16.126
	30	148.937	163.508	14.571
	40	188.761	203.819	15.058
	50	235.966	250.649	14.683
	60	294.958	312.069	17.111
T (mm)	1	151.986	162.826	10.84
	1.5	143.205	152.606	9.401
	2	140.301	153.022	12.721
	2.5	133.101	146.642	13.541
	3	132.117	146.516	14.399

where $\overline{\Delta R_i}$ is mean springback amount, ΔR_{ij} is the springback amount corresponding to the j -th level of parameter i , and $n_i = 5$ is the number of levels for each parameter.

The sample variance S_i^2 can be used to quantify the dispersion of the springback amount.

$$S_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\Delta R_{ij} - \overline{\Delta R_i})^2 \quad (51)$$

The relative influence weight W_i of parameter i can be obtained by normalizing its variance.

$$W_i = \frac{S_i^2}{\sum_{i=1}^m S_i^2} \times 100\% \quad (52)$$

where $m = 5$ is the total number of investigated influencing parameters.

3.3. Prediction process of WOA-Attention-BP model

Based on the analysis of the parameters affecting springback in Section 3.2.3, the key influencing factors are screened. Comprehensively considering the elastic recovery mechanism of the material during the unloading stage and the cumulative effect of plastic deformation during forming, the yield strength σ_s , elastic modulus E , and Poisson's ratio ν in the material constitutive relationship are listed as the core material parameters, and the wall thickness t is taken as the key structural dimension parameter. In the actual bending forming process, springback compensation is typically introduced in advance along the forming path to achieve the desired target shape. This compensation directly affects

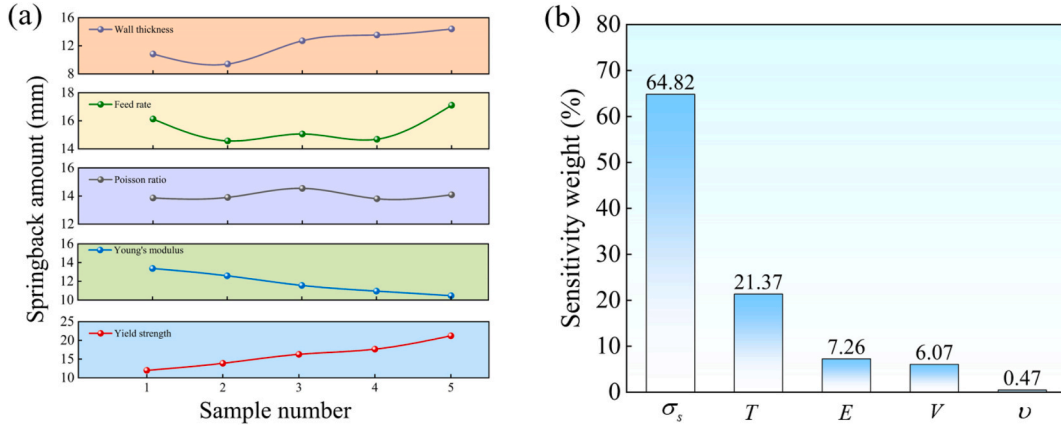


Fig. 19. Influence patterns and sensitivity analysis of various parameters on springback: (a) Variation trends of springback amount under different parameter levels; (b) Parameter sensitivity weight analysis.

Table 8
Sensitivity analysis results of influencing parameters on springback.

Parameters	Mean springback amount ΔR (mm)	Sample variance S_i^2	Influence weight W_i
σ_s	16.210	12.580	64.82%
T	12.180	4.147	21.37%
E	11.787	1.411	7.26%
V	15.509	1.178	6.07%
ν	14.041	0.091	0.47%

the motion trajectory of the bending die, thereby influencing the final springback result. Therefore, this study also incorporates the offset amount of the bending die after considering springback compensation into the system of influencing factors.

To systematically evaluate the accuracy of the WOA-Attention-BP intelligent prediction model proposed in this paper, this section designed the springback radius prediction process as shown in Fig. 20, and completed the model training and testing on the MATLAB platform under the environment of Intel Core i7-11800H processor.

3.3.1. Construction of learning samples

According to the FE simulation results, different material parameters, geometric parameters and process parameters have different degrees of influence on the springback, which can be described as:

$$\mathcal{D} = [U, \sigma_s, E, \nu, T] \quad (53)$$

where $\mathcal{D}$ represents the data information contained in the dataset.

In Section 3.2.3, 25 FE simulation samples G1 with different parameter settings were generated and used as part of the simulation training dataset. To further improve coverage of the multi-dimensional parameter space and ensure sufficient training sample size, an additional 25 parameter combinations G2 were generated using a five-factor, five-level $L_{25}(5^5)$ orthogonal array. As shown in Table 9, for each key factor, five engineering representative levels were selected, and the corresponding simulation results are shown in Fig. 21(a). Consequently, a total of 50 FE simulation samples G3 were obtained and used exclusively for model training.

3.3.2. Model performance evaluation

The model performance was evaluated using a multi-index comprehensive evaluation method, in which mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) are selected as quantitative evaluation indicators. Each indicator was calculated through Eqs. (54)–(56), respectively.

$$MAE = \frac{1}{N_s} \sum_{i=1}^{N_s} |\hat{R}_i - R_i^{\text{exp}}| \quad (54)$$

$$RMSE = \sqrt{\frac{1}{N_s} \sum_{i=1}^{N_s} (\hat{R}_i - R_i^{\text{exp}})^2} \quad (55)$$

$$MAPE = \frac{100\%}{N_s} \sum_{i=1}^{N_s} \left| \frac{\hat{R}_i - R_i^{\text{exp}}}{R_i^{\text{exp}}} \right| \quad (56)$$

where $\hat{R}_i$ is the predicted value of springback radius, R_i^{exp} is the experimental measured springback radius, N_s is the total number of test samples.

To ensure fairness in model comparison and reliability of experimental results, this study constructed four prediction models: BP, WOA-BP, Attention-BP and WOA-Attention-BP, all evaluated under a unified data partitioning and performance assessment scheme. The dataset comprises 50 FE simulation samples and 20 experiment samples. Considering the differences in uncertainty and error characteristics between simulation and experimental data, these two types of data were used separately during modeling. Specifically, 50 FE simulation samples were used for model training and optimal model selection, while the 20 experimental samples served as independent test data for evaluating model performance.

To avoid experimental randomness caused by single random data splits and enhance the stability of model selection results, this study conducted five-fold cross-validation based on 50 simulation samples. All simulation samples were evenly divided into five non-overlapping subsets. In each fold, four subsets (40 samples) were used as the training set, while the remaining one subset (10 samples) served as the validation set. This process was repeated five times, ensuring that each subset participates in validation exactly once. The model and hyperparameters were selected based on the average performance across the five folds. After determining the optimal hyperparameters, the four models were retrained comprehensively using all 50 FE simulation samples. The experimental samples were divided into an interpolation test set and an extrapolation test set based on whether the input parameters fall within the parameter coverage range of the simulation training data. The 10 sets of parameters that lie within the simulation training range formed the interpolation test set, which were used to evaluate the model's predictive performance. The remaining 10 sets of parameters outside the simulation training range constituted the extrapolation test set, which were used to assess the model's generalization capability beyond the training domain.

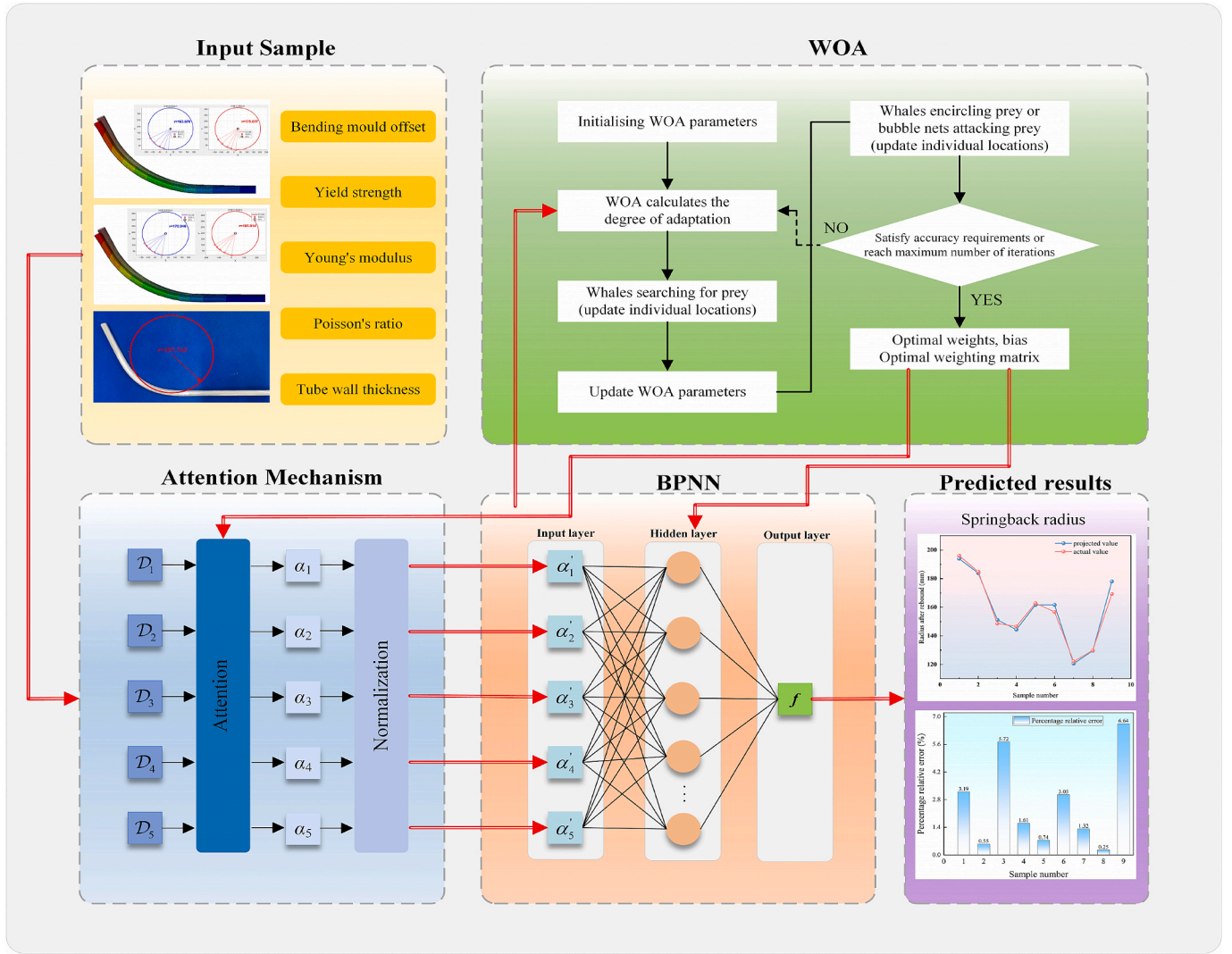


Fig. 20. Flowchart of the WOA-Attention-BP model for springback radius prediction.

To ensure consistent comparison conditions across all models, MSE was used as the loss function, with the hyperbolic tangent function employed as the activation function in the hidden layers. The training learning rate was set to 0.01, and the maximum number of iterations was 50. For models incorporating the WOA, the population size was uniformly set to 30. Additionally, to address overfitting commonly occurring in small-sample training scenarios, L2 regularization and early stopping were introduced during training, with the weight decay coefficient set to 0.001.

The five-fold cross-validation results of the WOA-Attention-BP model are shown in Table 10. The validation performance remains relatively stable across the five folds, with only limited fluctuations in MAE, RMSE and R^2 . Moreover, the training and validation errors remain at comparable levels, indicating that the model's performance is not dependent on specific data partitioning methods. The model's prediction performance on an independent test set is illustrated in Fig. 22. Fig. 22(a) shows the interpolation test set, where the predicted results show good consistency with the experimental results, achieving MAE, RMSE and MAPE of 3.13 mm, 3.52 mm and 2.09%, respectively. Fig. 22(b) presents the extrapolation test set, with corresponding evaluation metrics of 8.59 mm, 9.47 mm and 3.68%. These results indicate that despite the relatively small dataset size, no significant overfitting is observed. The model not only accurately predicts springback radius within known parameter ranges but also demonstrates certain extrapolation and generalization capabilities.

Fig. 23 shows the deviation between predicted and experimental results of springback radius on the test set using the WOA-BP, Attention-BP and BP

models. Compared to the single BP model, both the WOA algorithm and attention mechanism can improve prediction accuracy. As shown in Fig. 24 (a), after optimization by WOA, the MAE, RMSE and MAPE of the WOA-BP model decreased to 7.61 mm, 8.36 mm and 3.2%, respectively, with MAPE reduced by approximately 50% compared to the BP model. Further incorporating the attention mechanism can reduce MAE, RMSE and MAPE to 3.13 mm, 3.52 mm and 2.09%, respectively. These results clearly demonstrate that the WOA-Attention-BP model can focus on input features with greater influence on springback, while WOA can effectively strengthen the model's global optimization capability. The synergistic effect significantly improves the prediction accuracy and stability of the neural network in complex nonlinear regression tasks such as bending springback.

To further evaluate the reliability of the springback radius predicted by the WOA-Attention-BP model, the prediction uncertainty of the model was assessed based on the independent experimental validation results. For the i -th validation sample, the prediction residual is calculated as:

$$\varphi_i = \hat{R}_i - R_i^{\text{exp}} \quad (57)$$

Furthermore, the mean prediction residual can be calculated as:

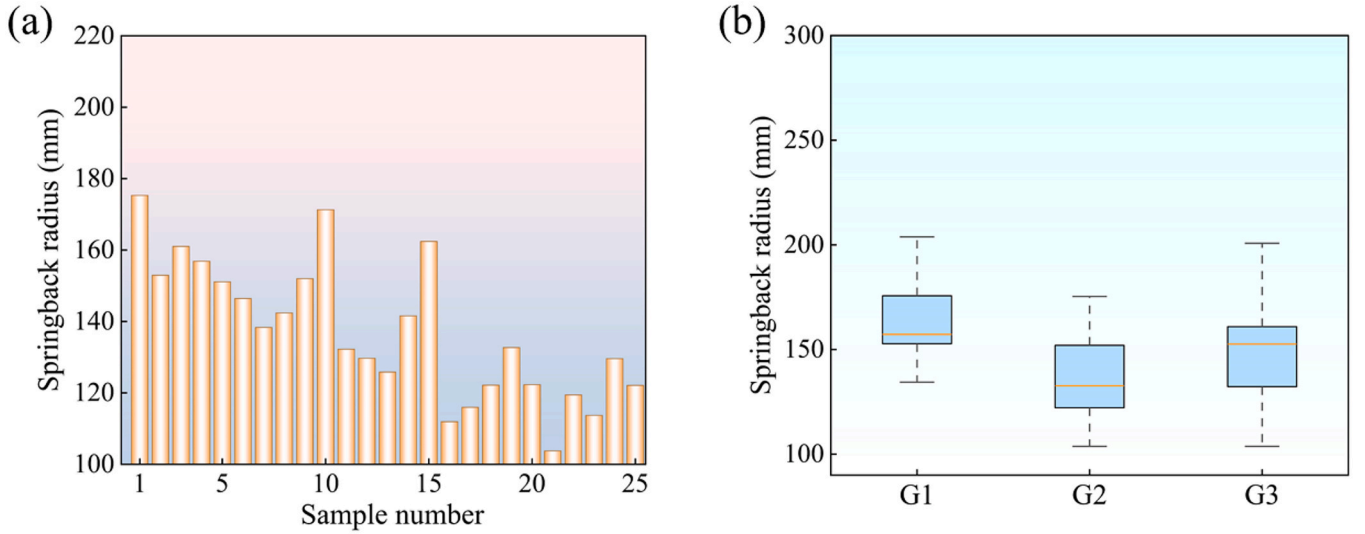
$$\bar{\varphi} = \frac{1}{N_s} \sum_{i=1}^{N_s} \varphi_i \quad (58)$$

where φ_i denotes the prediction residual, and $\bar{\varphi}$ represents the mean residual.

Table 9

Orthogonal design for simulation of free bending forming of tubes.

Levels	Factors	Bending die offset U (mm)	Yield strength σ_s (MPa)	Young's modulus E (GPa)	Poisson's ratio ν	Wall thickness T (mm)
1		10.2	180	62	0.31	1.0
2		10.2	220	64	0.32	1.5
3		10.2	260	66	0.33	2.0
4		10.2	300	68	0.34	2.5
5		10.2	340	70	0.35	3.0
6		10.71	180	68	0.33	1.5
7		10.71	220	70	0.34	2.0
8		10.71	260	62	0.35	2.5
9		10.71	300	64	0.31	3.0
10		10.71	340	66	0.32	1.0
11		11.22	180	64	0.35	2.0
12		11.22	220	66	0.31	2.5
13		11.22	260	68	0.32	3.0
14		11.22	300	70	0.33	1.0
15		11.22	340	62	0.34	1.5
16		11.73	180	70	0.32	2.5
17		11.73	220	62	0.33	3.0
18		11.73	260	64	0.34	1.0
19		11.73	300	66	0.35	1.5
20		11.73	340	68	0.31	2.0
21		12.24	180	66	0.34	3.0
22		12.24	220	68	0.35	1.0
23		12.24	260	70	0.31	1.5
24		12.24	300	62	0.32	2.0
25		12.24	340	64	0.33	2.5

**Fig. 21.** Springback results obtained from tube free bending simulation: (a) Orthogonal experiment simulation results; (b) Springback radius distribution, G1 is single-factor simulation samples, G2 is orthogonal experimental simulation samples, and G3 is all simulation samples.**Table 10**

Five-fold cross-validation results of the WOA-Attention-BP model.

Fold	MAE (mm)	RMSE (mm)	MAPE (%)	R^2
1	3.38	4.31	2.21	0.97
2	5.52	6.80	3.62	0.92
3	6.30	9.19	4.16	0.88
4	5.03	5.14	2.67	0.96
5	6.42	8.46	4.05	0.89
Mean $\pm$ SD	5.33 $\pm$ 1.23	6.78 $\pm$ 2.09	3.34 $\pm$ 0.86	0.92 $\pm$ 0.04

By combining Eqs. (57), (58) and (55), the standard deviation of the prediction residuals can be obtained as:

$$S_\varphi = \sqrt{\frac{1}{N_s - 1} \sum_{i=1}^{N_s} (\varphi_i - \bar{\varphi})^2} = \sqrt{\frac{N_s}{N_s - 1} (RMSE^2 - \bar{\varphi}^2)} \quad (59)$$

where RMSE is the overall deviation between the predicted results and the experimental measurements.

Since the experimentally measured springback radius contains measurement uncertainty, and the material properties and process parameters may also introduce uncertainties during the training process of the WOA-Attention-BP model, the prediction residual variance can be further expressed as:

$$S_\varphi^2 = u_m^2 + u_c^2 \quad (60)$$

where u_m represents the standard prediction uncertainty of the WOA-Attention-BP model itself, and u_c represents the combined standard uncertainty of the experimental measurement.

As shown in Fig. 23(a), the prediction residuals were calculated based on 10 independent experimental validation samples. The mean residual was obtained as $\bar{\varphi} = 0.47$ mm, which is relatively small

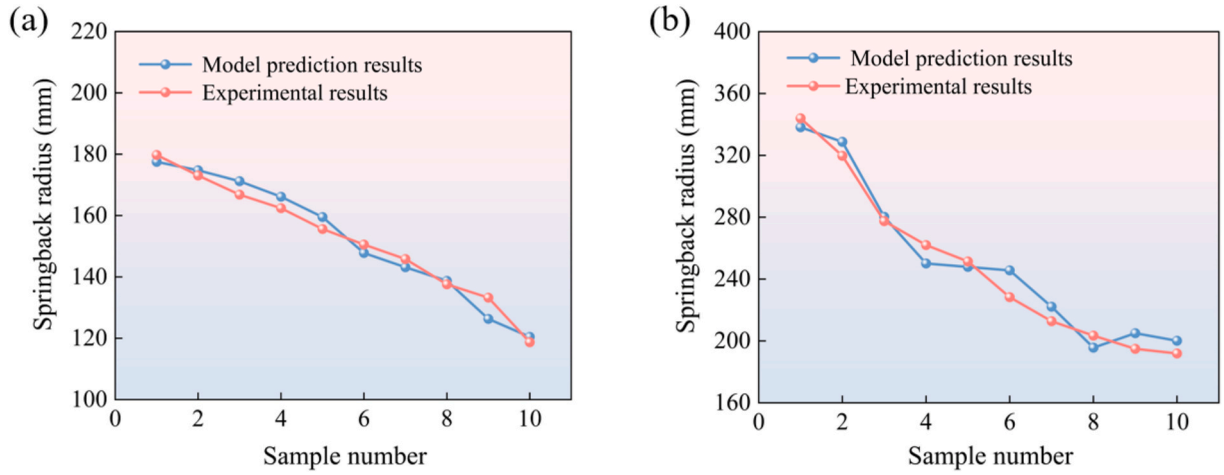


Fig. 22. Comparison between WOA-Attention-BP model predictions and experimental springback radius: (a) Interpolation test set (parameters within training range); (b) Extrapolation test set (parameters beyond training range).

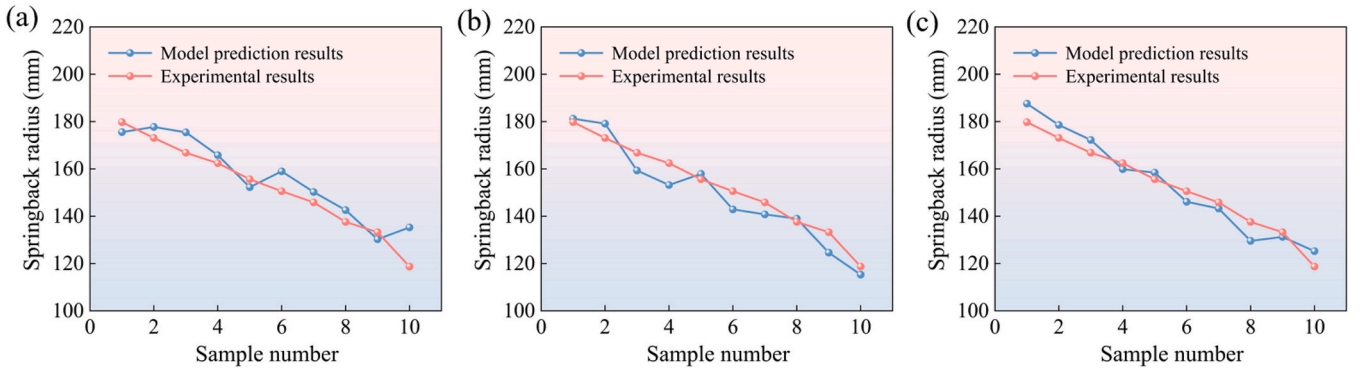


Fig. 23. Comparison of springback radius predictions under different models: (a) BP model; (b) Attention-BP model; (c) WOA-BP model.

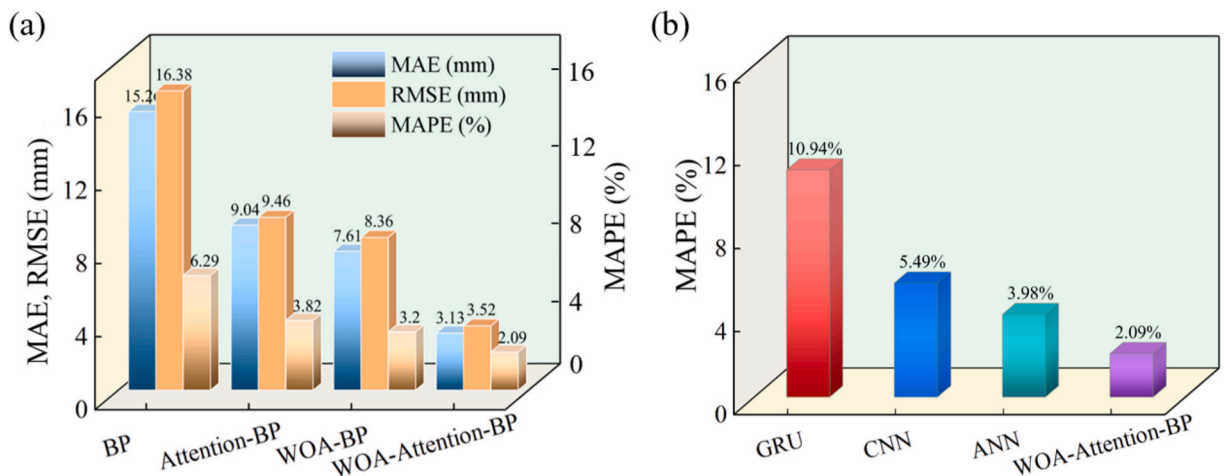


Fig. 24. Performance evaluation of springback prediction for different models: (a) The impact of WOA algorithm and Attention mechanism on prediction accuracy; (b) Prediction errors of various machine learning models.

compared with $RMSE = 3.52$ mm. Moreover, the prediction residuals fluctuate around zero, indicating that the model does not exhibit obvious systematic overestimation or underestimation tendencies.

By substituting the experimentally obtained combined standard uncertainty $u_c = 0.43$ mm from Section 2.2 into Eq. (60), the standard prediction uncertainty of the WOA-Attention-BP model can be further calculated as 3.67 mm. When the coverage factor is taken as $k = 2$, the

expanded uncertainty is calculated as 7.34 mm, corresponding to an approximately 95% confidence level. It should be noted that during the prediction process, the effects of the input material and process parameters on the springback radius have already been incorporated into the final prediction residual through the training samples and the nonlinear mapping capability of the WOA-Attention-BP model. Therefore, this study evaluates the overall prediction uncertainty of the model

Table 11
Network architectures and main parameters of different comparison models.

Model	Basic network architecture	Key structural parameters	Special modules and parameters
ANN	Single-hidden-layer fully connected neural network	Number of hidden-layer neurons: 4	L_2 regularization coefficient: 0.001
CNN	One-dimensional convolutional neural network	Kernel size: 3; number of convolutional filters: 4	ReLU activation, and global average pooling; L_2 regularization coefficient: 0.001
GRU	Single-layer gated recurrent unit network	Number of GRU hidden units: 4	The hidden state at the final time step is used as the network output; L_2 regularization coefficient: 0.001

based on independent experimental validation residuals, rather than separately decomposing the uncertainty contribution of individual input parameters.

To further demonstrate the superiority of WOA-Attention-BP under small sample conditions, several classical machine learning models were employed for springback radius prediction. To ensure a fair comparison among different models, WOA-Attention-BP, ANN, CNN and GRU models all adopted the same data partitioning method, normalization technique, maximum number of training epochs and loss function. The specific network architectures and hyperparameter settings for each model are shown in Table 11.

Fig. 24(b) shows the prediction accuracy of several classical machine learning models on this dataset. Among the compared models, GRU, CNN, and ANN achieved MAPE values of 10.94%, 5.49%, and 3.98%, respectively, while the proposed WOA-Attention-BP model achieved the lowest MAPE of 2.09%. This further indicates that under small sample conditions, the advantages of deep networks such as GRU and CNN are difficult to fully realize, likely owing to insufficient feature learning or overfitting issues. In contrast, the synergy between global parameter searching of WOA and adaptive critical feature screening of the attention mechanism fundamentally accounts for the prominent accuracy improvement of the WOA-Attention-BP model.

4. Conclusions

This study addresses the issues of high uncertainty in measuring springback for tube free bending and the difficulty in balancing accuracy and efficiency of prediction models. Taking Al6061 aluminum alloy tubes as the subject, a metrologically characterized method for measuring springback radius was proposed, and the estimation of springback radius was achieved by integrating analytical models, finite element simulation, and a WOA-Attention-BP prediction model. The main conclusions are as follows:

- (1) A metrologically characterized method for measuring the springback radius of tube free bending was proposed. A two-stage circle fitting algorithm was established by combining Kasa algebraic initialization with Gauss-Newton optimization based on geometric distance. The influence of coordinate perturbations on the fitted radius was quantified through a radius error amplification factor, which significantly decreases as the fitted arc angle increases.
- (2) A quantitative analysis was conducted on the uncertainty propagation chain established for 3D laser scanning, point-cloud processing, centerline extraction, and circle fitting. The results indicate that the method has a combined standard uncertainty of 0.43 mm, an expanded uncertainty of 0.86 mm, and a repeatability standard deviation of 0.63 mm. Compared with validated finite element model results, the average relative deviation is 2.57%, confirming the accuracy and repeatability of the proposed method in measuring springback radius.
- (3) An analytical springback model considering neutral layer shifting was established, which can calculate the position of the elastoplastic

boundary, the neutral layer position and the forming bending moment under a given bending radius, enabling estimation of the springback radius. Within the relative bending radius range of $R/D = 5\text{--}8$, the relative deviations range from 6.03% to 6.80%.

(4) A hybrid WOA-Attention-BP framework was developed to achieve rapid springback radius prediction under complex process conditions. On the validation dataset, the proposed model achieved MAE, RMSE, and MAPE values of 3.13 mm, 3.52 mm, and 2.09%, respectively. Furthermore, by integrating measurement uncertainty propagation with model residual analysis, the standard prediction uncertainty of the model was quantified as 3.67 mm.

In summary, this study establishes a comprehensive framework integrating accurate measurement, uncertainty evaluation, and intelligent prediction of springback radius in tube free bending, providing a theoretical foundation for springback measurement and compensation. Future work will investigate multi-material transfer learning approaches to extend model applicability, and integrate real-time force and displacement feedback to achieve online model updating and adaptive optimization.

AI usage statement

The authors did not use generative AI or AI-assisted technologies in the writing, analysis, or preparation of this manuscript.

CRediT authorship contribution statement

Shuai Zhang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chengyu Fu:** Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Writing – review & editing. **Zhenming Yue:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Hao Qu:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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